Pure Mathematics for Engineers and Scientists

Anthony Daniell

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Preface

The basic aim is to provide exposure to a solid foundation of mathematics based on proofs. The exit goals would be:

-Readers will be able to write valid proofs using the standard techniques of modern mathematics.

-Readers will be able to read and understand proofs at some reasonable level.

-Readers will have working knowledge of broad foundations of mathematics, and be able to confidently take dedicated courses in each area, or pursue independent studies.

From my experience, I certainly never got this stuff explicitly, but on retrospect, it certainly seems like it would help.

My original concept was that the course would be targeted from someone maybe at the junior or senior level in physics/engineering. So, the typical prerequisite would be calculus, maybe differential equations, and experience applying this math to physics, engineering problems like in mechanics, electricity and magnetism, etc. That's essentially my background, so that's why I thought of this cohort.

The main reason is that this person, in my concept, would have some context and exposure to different math, and may be better motivated to wanting to understand the underpinnings. Also, they may be more interested to branch out into the other topics on the list as these will likely be important as they take more advanced classes in their careers.

However, that being said, I think the main prerequisite is being interested in pure mathematics, perhaps with some future application in mind, but not necessarily.

Topics to be covered:

Mathematical Foundations - Mathematical logic, basic notation, axiomatic set theory (Zermelo-Fraenkel), Peano Axioms of Arithmetic, Gödel theorems.

Methods of Proof: Induction, Contradiction, Direct Proofs. Examples of classic proofs - infinite primes, irrationality of square root of two, etc. Plenty of practice examples.

Modern Algebra: Groups, Rings, Fields, Transformations, Representation Theory, Modules, Galois Theory

Real Analysis - Study of real numbers, including construction by Dedekind Cuts etc, point set topology, continuity, metric spaces (Basically a compressed version of Rudin or equivalent).

Complex Analysis - Study of functions of a complex variable with connections to other areas of mathematics including number theory. Riemman Zeta Function and Riemann Hypothesis.

Non-euclidean geometry and topology: Various spaces and properties. Invariants, knot theory, homotopy, homology.

Number theory:

Category theory:

Computational complexity theory: Finite Automata, lambda calculus, NP and other spaces.

Contents

Chapter 1

Mathematical Foundations

The objective of this chapter is to present some basic concepts that underlie much of the remaining material in this book.

1.1 Basic Notation

The reader may be familiar with some if not all of the following notation, but it is presented here for concreteness and review.

- $\forall$ is read as "for all"
- $\exists$ is read as "exists"
- $\subset$ is read as "subset of"
- $\in$ is read as "in"
- $\mathbb{Z}$ is the set of integers: $\ldots, -2, -1, 0, 1, 2, 3, \ldots$
- $\mathbb{N}$ is the set of natural numbers: $0, 1, 2, 3, \ldots$ (note: Sometimes these start with 0, and sometimes with 1. Usually doesn't matter to the argument in question, but the reader should be aware.)
- $\mathbb{Q}$ is the set of rational numbers. These are numbers of the form $\frac{p}{q}$ where p, q are integers, and $q \neq 0$
- $\mathbb{R}$ is the set of real numbers. These will later be defined precisely in terms of *Dedekind Cuts*, but for now we'll just use the informal notion of all the points on a coordinate line. However, hopefully the reader can see this is not good enough for precise work.
- We use curly braces to indicate sets in general. For example, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, 3\ldots\}$ defines the set of integers using the curly braces and examples. It is hoped the reader is able to abstract that the dots to the left imply the list has

no smallest negative value and the dots to the right imply that there is no largest positive value.

- Set builder notation: There are other ways to specify the set within the braces. For example, we can indicate the property that each member of the set has: $S = \{p|p = 2m + 1, m \in \mathbb{Z}\}$. This is read as the set S consists of those values p of the form $2m+1$ where m is an integer. In other words, S the set of odd integers. The vertical bar in the braces is read as "such that" and the comma is read as "and."

Exercise
Write the definition of the set of rational numbers using set builder notation.
Solution
$\mathbb{Q} = \left\{r|r = \frac{p}{q}, p, q \in \mathbb{Z}, q \neq 0\right\}$

1.2 Mathematical Logic

It should be remarked that a good portion of this section (as well as some others) was strongly inspired by the YouTube video series "Math Major Basics" by MathDoctorBob (https://www.youtube.com/playlist?list=PLF2DF6C3C8015DF5F). We cover some key aspects here. It is very helpful to understand *Truth Tables* in this context. For those who have some experience with digital logic or computer programming, much of this will be quite familiar.

We have the following logical relations, with symbols, defined by the truth tables shown below: Negation ($\neg$), Conjunction/And ($\wedge$ - note that this symbol looks like the outline of the letter "A", to help you remember "And"), Disjunction/or ($\vee$), Implication ($\Longrightarrow$), If and Only If ($\Longleftrightarrow$):

Negation:

A	$\neg A$
T	F
F	T

Conjunction/And:

A	$\wedge$	B
T	T	T
T	F	F
F	F	T
F	F	F

Disjunction/Or:

A	$\vee$	B
T	T	T
T	T	F
F	T	T
F	F	F

Implication:

A	$\Longrightarrow$	B
T	T	T
T	F	F
F	T	T
F	T	F

If and only if (Iff):

A	$\Longleftrightarrow$	B
T	T	T
T	F	F
F	F	T
F	T	F

In the above, A and B are statements which can be either true (T) or false (F). The result of the logical operation is shown in the column under the particular symbol. For example, if A is true, and B is true, then $A \wedge B$ is true. However, either either A or B is false (or both) then $A \wedge B$ is false. It is important to refer back to the truth table definitions of these logical operations to avoid confusion.

Sometimes, implication is called "if-then" as in "If A then B." However, note that it is possible for A to be false and B to be true, and the implication $A \Longrightarrow B$ to be true. This is somewhat at odds with how this if-then might be expressed in common usage: How could something false imply something true? This seems confusing. Therefore, always refer to the truth table for clarity on this point.

Here is an illustration regarding implication ($\Longrightarrow$).

Let $A = (a > 5), B = (a > 3)$ where a is some integer. Now consider the following cases:
If $a = 6, (6 > 5) = T, (6 > 3) = T$
If $a = 4, (4 > 5) = F, (4 > 3) = T$
If $a = 2, (2 > 5) = F, (2 > 3) = F$

If we compare this to the truth table for implication, we see that implication is always true. Note that with this particular definition of the statements A and B we cannot have a case where A is true and B is false.

Exercise Show $A \Longleftrightarrow B$ is equivalent to $(A \Longrightarrow B) \wedge (B \Longrightarrow A)$.

Solution

$(A$	$\Longrightarrow$	$B)$	$\wedge$	$(B$	$\Longrightarrow$	$A)$
T	T	T	T	T	T	T
T	F	F	F	F	T	T
F	T	T	F	T	F	F
F	T	F	T	F	T	F

We now consider some important terminology:

- *Tautology*: Any statement that is true for all values of its components is a tautology.
- *Contradiction*: Any statement that is false for all values of its components is a contradiction. The negation of a contradiction is a tautology and the negation of a tautology is a contradiction.
- *Logically Equivalent*: When $A \iff B$ is a tautology. This means A and B have the same truth table. The notation for this is $A \equiv B$.

We illustrate logical equivalence by considering "De Morgan's Laws" for disjunction (or) and conjunction (and). De Morgan's Law's are useful for manipulating logical expressions:

$$\neg (A \vee B) \equiv (\neg A) \wedge (\neg B)$$
$$\neg (A \wedge B) \equiv (\neg A) \vee (\neg B)$$

In words, the first law says that the negation ($\neg$) of a disjunction ($A \vee B$) is logically equivalent to the conjunction of the negations of the individual statements A and B. The second law is similar, except we exchange conjunction for disjunction and vice-versa.

Exercise Verify the De Morgan law for disjunction by showing the logical equivalence of the left and right hand sides using a truth table:
Solution

$\neg$	$(A$	$\vee$	$B)$	$\iff$	$(\neg$	$A)$	$\wedge$	$(\neg$	$B)$
F	T	T	T	T	F	T	F	F	T
F	T	T	F	T	F	T	F	T	F
F	F	T	T	T	T	F	F	F	T
T	F	F	F	T	T	F	T	T	F

We note that for any value of statements A and B, the values under the column for $\iff$ are all true, indicating that both expressions $\neg (A \vee B)$, $(\neg A) \wedge (\neg B)$ have the same truth tables, as required.

Another useful logical equivalence is the following "contrapositive" form of implication, which we explore in the next exercise:

Exercise Show that $(A \Longrightarrow B) \iff (\neg B \Longrightarrow \neg A)$ is a tautology, i.e. this means that $A \Longrightarrow B$ is logically equivalent (LE) to $\neg B \Longrightarrow \neg A$.

Solution

$(A$	$\Longrightarrow$	$B)$	$\iff$	$(\neg$	B	$\Longrightarrow$	$\neg$	$A)$
T	T	T	T	F	T	T	F	T
T	F	F	T	T	F	F	F	T
F	T	T	T	F	T	T	T	F
F	T	F	T	T	F	T	T	F

To show a situation where expressions are not logicaly equivalent, the following exercise may be considered:

Exercise Show $A \Longrightarrow B$ is not logically equivalent to $B \Longrightarrow A$

Solution

$(A$	$\Longrightarrow$	$B)$	$\iff$	$(B$	$\Longrightarrow$	$A)$
T	T	T	T	T	T	T
T	F	F	F	F	T	T
F	T	T	F	T	F	F
F	T	F	T	F	T	F

Notice the middle column $\iff$ has false entries, so we don't have logical equivalency.

Another item of terminology is *Logical Implication*. Logical implication is when the expression $A \Longrightarrow B$ is a tautology. If the statements A and B are such that we only get true statements from $A \Longrightarrow B$, then A logically implies **(LI)** B. This was the situation with the earlier illustration where we had defined statements A and B such that $A = (a > 5), B = (a > 3)$ where a is some integer.

If A is logically equivalent (LE) to B, then A LI B and B LI A.

Exercise: Show $A \wedge (A \Longrightarrow B)$ logically implies B. This is called "Modus Ponens."

Solution

$(A$	$\wedge$	$(A$	$\Longrightarrow$	$B))$	$\Longrightarrow$	B
T	T	T	T	T	T	T
T	F	T	F	F	T	F
F	F	F	T	T	T	T
F	F	F	T	F	T	F

1.3 Zermelo-Fraenkel Axiomatic Set Theory

Set theory is at the foundation of mathematics. The reader should have some awareness of the axioms of set theory, although in practice it seems that one rarely sees a mathematical proof that explicitly cites these axioms. These axioms are referred to as the Zermelo-Frankel Axioms.

1.3.1 The Zermelo-Fraenkel Axioms

The following so-called Zermelo-Fraenkel Axioms of set theory are presented to help provide the reader a more comprehensive picture of mathematics foundations. This section is inspired by the set of YouTube lectures by Richard Borcherds (ZF Axioms)
(`https://www.youtube.com/playlist?list=PL8yHsr3EFj52EKVgPi-p50fRP2_SbG2oi`)
and also the book *Axiomatic Set Theory* by Paul Bernays. The definitions are adapted from Bernays.

We note that in the axioms below, the notion of *set* is undefined, as is the inclusion relation, $\in$. Furthermore, all elements or members of sets are sets themselves.

Axiom of Extensionality
If $s \subseteq t$ and $t \subseteq s$, then $s = t$ where s, t are sets and $\subseteq$ means *subset*.

We define subset by the following: If s and t are sets such that, for all x, $x \in s$ implies $x \in t$, s is called a subset of t, where x is a set element or member. Occasionally, we might use the notation $s \subset t$ to indicate a *proper subset* whereby there is an element in t that is not in s.

To recap: The axiom states what it means for two sets to be equal - namely they contain the same elements.

Axiom of Foundation
Every non-empty set s contains an element t such that s and t have no common element.

This axiom prevents an element of a set having an element which appears in the original set. For example, the set $s = \{s\}$ is not allowed.

Axiom of Pairing
For any two different sets a and b, the pair $\{a, b\}$, or $\{b, a\}$, exists.

Axiom of Union
For any set s which contains at least two elements, there exists the set whose elements are the elements of the elements of s.

This set is called the union of the elements of s and is denoted by $\cup s$.

Axiom of Infinity
There exists at least one set W with the properties:

- a) $\emptyset \in W$, where $\emptyset$ refers to the "empty set." - the set with no elements.
- b) if $x \in W$, then $\{x\} \in W$.

The interpretation of the above is that a set exists with a non-ending sequence of elements. For example, if we think of the infinite sequence of non-negative integers being encoded as follows: $0 = \emptyset, 1 = \{0\}, 2 = \{1\}, \ldots$, then we can have the corresponding infinite set $S = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \ldots\}$ which is justified by the axiom.

Axiom of Power-set
For any set s, there exists the set whose elements are all subsets of s.

To take an example, consider the set $s = \{1, 2, 3\}$. Then the power-set is $P = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}, \emptyset\}$. The number of elements, also called the *cardinality*, in this case is given by 2^N, where N is the number of elements in the original set s. Thus, we have $2^{N=3} = 8$. We note that the empty set, $\emptyset$, is considered a subset of every set.

Axiom of Subsets (Note: Borcherds refers to this as "separation")
For any set s and any predicate P which is meaningful for all elements of s, there exists the set y that contains just those elements x of s which satisfy the predicate P.

For example, let $s = \{1, 2, 3, 4, 5\}$ and the predicate $P(x)$ be true whenever x is even. Then, we have the set $y = \{x \in s | P(x) true\} = \{2, 4\}$.

Axiom of Replacement
For every set s and every single-valued function f of one argument which is defined for all the elements of s, there exists the set that contains all $f(x)$ with $x \in s$.

Axiom of Choice
For every disjointed set t for which $\emptyset \notin t$, the Cartesian product $\mathfrak{P}t$ differs from the $\emptyset$.

We define a *disjointed set* by the following: If a set s contains at least two elements, and any two elements of s are mutually exclusive (i.e. these elements have no common elements themselves).

Theorem The *Cartesian product* of the elements of a disjointed set t, denoted $\mathfrak{P}t$ is the set whose elements are the sets which contain a single element from each element of t. If $a, b, \ldots$ are the elements of t, the Cartesian product is also denoted by $a \times b \times \ldots$. *Note*: For those interested, this theorem is proved in Bernays.

To illustrate the applications of the above axioms to a proof, consider the following exercise.

Exercise Complete the proof of the following theorem (adapted from Bernays):

Theorem For every non-empty set t, there exists the set whose elements are common to all elements of t.
This set is called the *intersection* of the elements of t and is denoted by $\cap t$.

Proof: By the Axiom of Union, $s = \cup t$ exists and contains the elements of the elements of t. The predicate "x is contained in each element of t" defines, by the Axiom of **fill in the blank**, a subset of s which is the intersection $\cap t$.

Solution
Axiom of **Subsets**

The following exercise explores the Cartesian product.

Exercise According to Bernays, for a disjointed set t, one has the Cartesian product, $\mathfrak{P}t = \emptyset$ if $\emptyset \in t$. Prove this using the Cartesian Product Theorem stated above in Axiom of Choice section.

Solution According to the theorem, the Cartesian product is the set whose elements are the sets which contain a single element from each element of t. Since $\emptyset \in t$ and contains no elements, the Cartesian product must be empty since there is no set possible that contains the required one element from $\emptyset$.

1.3.2 Set Operations

In this section, we present some set operations that are useful in much of the remainder of this book. The reader may be familiar with some of the material, although perhaps not necessarily in the formalism presented. Concepts and proofs in mathematics are very frequently expressed in terms of set language to be precise, so utility with set operations is important. For example, in later sections, we will discuss algebraic structures such as "groups" which will be defined in terms of sets with certain properties.

The emphasis here will be to continue the development following the axiomatic approach, largely adapted from *Axiomatic Set Theory* by Patrick Suppes. Hopefully, this will illustrate some of the notation and concepts we have been developing in the previous sections.

Binary Relations

Definition: *R is a binary relation* $\iff (\forall x)(x \in R \Longrightarrow (\exists y)(\exists z)(x = \langle y, z\rangle))$.

We note that in the above, $\langle y, z\rangle$ is called an *ordered pair* and is formally defined in set notation as $\langle y, z\rangle = \{\{y\}, \{y, z\}\}$. Note how the ordering is encoded. So we have that a relation is a set of ordered pairs.

The definition is read as "R is a binary relation if and only if for all x if x is in R, then there exists a y and there exists a z such that x is equal to the ordered pair $\langle y, z\rangle$."

It should be remembered that the phrases "if and only if" and the "if ... then" are used to refer to the truth tables for the corresponding symbols and not the casual english meaning. If desired, we could express the definition by using a truth table as we did previously to be completely clear.

Sometimes the following notation is used discussing relations: xRy. This is formally defined as $xRy \iff \langle x, y\rangle \in R$.

Consider the following example set which is a relation:

$$R = \{\langle 1, 2\rangle, \langle 2, 1\rangle, \langle 1, 3\rangle, \langle 3, 1\rangle, \langle 2, 3\rangle, \langle 3, 2\rangle\}.$$

We have utilized the angle bracket notation for ordered pairs for simplicity. Observe that this relation, R, has some special properties. To begin with, this relation is called *symmetric* because if xRy is true then we have yRx true. For example, $\langle 1, 2\rangle \in R$ and $\langle 2, 1\rangle \in R$.

Exercise For the relation R defined above, verify that it is symmetric by explicitly listing the ordered pairs in question and their corresponding symmetric partners.

Solution
$\langle 1, 2\rangle, \langle 2, 1\rangle$
$\langle 2, 1\rangle, \langle 1, 2\rangle$
$\langle 1, 3\rangle, \langle 3, 1\rangle$
$\langle 3, 1\rangle, \langle 1, 3\rangle$
$\langle 2, 3\rangle, \langle 3, 2\rangle$
$\langle 3, 2\rangle, \langle 2, 3\rangle$

We consider as an exercise, the proof that $\emptyset$ is a relation.

Exercise Prove that $\emptyset$ (the empty set) is a relation by appealing to the definition of relation given earlier.
Solution
Recalling the definition: *R is a binary relation* $\iff (\forall x)(x \in R \Longrightarrow (\exists y)(\exists z)(x =$

$\langle y, z\rangle))$, we substitute $\emptyset$ for R and obtain:

$$\emptyset \textit{ is a binary relation } \iff (\forall x)(x \in \emptyset \implies (\exists y)(\exists z)(x = \langle y, z\rangle)).$$

Now, since $x \in \emptyset$ is false (the empty set contains no members), the statement $(x \in \emptyset \implies (\exists y)(\exists z)(x = \langle y, z\rangle))$ is true. Recall the truth table for $\implies$ whereby if the first term is false, the entire statement is true no matter what the second term is. Therefore, we have shown the $\emptyset$ is a relation because it satisfies the definition.

Ordering Relations

In the previous section, we introduced the idea of a *symmetric* relation. We now present a more complete list of such definitions:

- *R is reflexive in A* $\iff (\forall x)(x \in A \implies xRx)$.
- *R is irreflexive in A* $\iff (\forall x)(x \in A \implies \neg(xRx))$.
- *R is symmetric in A* $\iff (\forall x)(\forall y)(x, y \in A \wedge xRy \implies yRx)$.
- *R is asymmetric in A* $\iff (\forall x)(\forall y)(x, y \in A \wedge xRy \implies \neg(yRx))$.
- *R is antisymmetric in A* $\iff (\forall x)(\forall y)(x, y \in A \wedge xRy \wedge yRx \implies x = y)$.
- *R is transitive in A* $\iff (\forall x)(\forall y)(\forall z)(x, y, z \in A \wedge xRy \wedge yRz \implies xRz)$.
- *R is connected in A* $\iff (\forall x)(\forall y)(x, y \in A \wedge x \neq y \implies xRy \vee yRx)$.
- *R is strongly connected in A* $\iff (\forall x)(\forall y)(x, y \in A \implies xRy \vee yRx)$.

A couple of items regarding notation. The notation $x, y \in A$ means $x \in A \wedge y \in A$. Also, by convention, the $\wedge$ and $\vee$ operations are performed before $\implies$ which saves writing extra parentheses in the definitions. For example, if we explicitly included the parentheses in the definition of symmetric to indicate order of operations, it would appear as $A(\forall x)(\forall y)((x, y \in A \wedge xRy) \implies yRx)$.

If we consider the definition of *symmetric* above, we see how this corresponds to the example symmetric relation in the previous section. The reader should be cautioned to carefully observe the difference between <u>*a*</u>*symmetric* and <u>*anti*</u>*symmetric* relations.

Exercise Prove that the relation from the previous section is irreflexive by appealing to the definition. Define the set $A = \{1, 2, 3\}$ for purposes of this exercise.

Solution

Recalling R:

$$R = \{\langle 1,2\rangle, \langle 2,1\rangle, \langle 1,3\rangle, \langle 3,1\rangle, \langle 2,3\rangle, \langle 3,2\rangle\}.$$

we see that $\forall x \in A$, namely $1, 2, 3$, we are missing the corresponding ordered pairs $\langle 1,1\rangle, \langle 2,2\rangle, \langle 3,3\rangle$ in the relation R. Now if we consider the definition of *irreflexive* we have: *R is irreflexive in A* $\iff (\forall x)(x \in A \implies \neg(xRx))$. Clearly, xRx is false for any choice of $x \in A$, thus we have have $\neg(xRx)$ true for all choices of $x \in A$ and we see that R is therefore indeed *irreflexive*.

Exercise Show by counterexample, that the relation from the previous section is not *transitive*. Once again, consider the set $A = \{1, 2, 3\}$ for purposes of this exercise.

Solution

Recalling the definition: *R is transitive in A* $\iff (\forall x)(\forall y)(\forall z)(x, y, z \in A \wedge xRy \wedge yRz \implies xRz)$, we see that if we can find an instance where $x, y, z \in A \wedge xRy \wedge yRz \implies xRz$ is false, we are done. Consider $1R2$ and $2R1$ which are both true. However, $1R1$ is false because $\langle 1,1\rangle \notin R$. Therefore, we conclude that R is not transitive.

The following two definitions are important:

- *R is a partial ordering of A* $\iff$ *R is reflexive, anti-symmetric and transitive in A.*
- *R well-orders A* $\iff$ *R is connected in A* $\wedge (\forall B)(B \subseteq A \wedge B \neq \emptyset \implies$ *B has an R-minimal element*$)$.

The definition of *R-minimal element* mentioned above is as follows: *x is an R-minimal element of A* $\iff x \in A \wedge (\forall y)(y \in A \implies \neg(yRx))$.

Let us examine the above two ordering definitions with some examples. Firstly, let us consider *partial ordering*. We will take $A = \{1, 2, 3\}$ and construct an appropriate relation. We can begin with taking all possible ordered pairs which gives us the following provisional relation:

$$R = \{\langle 1,1\rangle, \langle 1,2\rangle, \langle 1,3\rangle, \langle 2,1\rangle, \langle 2,2\rangle, \langle 2,3\rangle, \langle 3,1\rangle, \langle 3,2\rangle, \langle 3,3\rangle\}$$

The above relation is symmetric, because it contains $\langle 1,1\rangle, \langle 2,2\rangle, \langle 3,3\rangle$. The anti-symmetric requirement restricts the ordered pair partners. For example, if we have $\langle 1,2\rangle$ we cannot have $\langle 2,1\rangle$. So, let us remove those ordered pairs where the second element is smaller than the first:

$$R = \{\langle 1,1\rangle, \langle 1,2\rangle, \langle 1,3\rangle, \langle 2,2\rangle, \langle 2,3\rangle, \langle 3,3\rangle\}$$

Lastly, we need to verify transitivity:
$1R1 \wedge 1R1 \implies 1R1$
$1R1 \wedge 1R2 \implies 1R2$

$1R1 \wedge 1R3 \Longrightarrow 1R3$
$1R2 \wedge 2R2 \Longrightarrow 1R2$
$1R2 \wedge 2R3 \Longrightarrow 1R3$
$1R3 \wedge 3R3 \Longrightarrow 1R3$
$2R2 \wedge 2R2 \Longrightarrow 2R2$
$2R2 \wedge 2R3 \Longrightarrow 2R3$
$2R3 \wedge 3R3 \Longrightarrow 2R3$
$3R3 \wedge 3R3 \Longrightarrow 3R3$

So, our revised R is a partial ordering of the set A. When dealing with real numbers, the symbol $\leq$ (less than or equal to) represents a partial ordering relation. The relation R we constructed above parallels this.

Now, let us do a similar construction to obtain a well-ordering. Again, consider the set $A = \{1, 2, 3\}$ and let us construct a relation R that satisfies the definition. To begin with, we can take as provisional all possible ordered pairs:

$$R = \{\langle 1,1\rangle, \langle 1,2\rangle, \langle 1,3\rangle, \langle 2,1\rangle, \langle 2,2\rangle, \langle 2,3\rangle, \langle 3,1\rangle, \langle 3,2\rangle, \langle 3,3\rangle\}$$

Now, we first need to make sure that R is *connected* in A by appealing to the definition of connected: R *is connected in* $A \iff (\forall x)(\forall y)(x, y \in A \wedge x \neq y \Longrightarrow xRy \vee yRx)$. We have as members of R, $\langle 1,2\rangle, \langle 1,3\rangle, \langle 2,3\rangle$, so R is connected.

Next, we need to verify the second part of the definition: $(\forall B)(B \subseteq A \wedge B \neq \emptyset \Longrightarrow B$ *has an* R*-minimal element*). There are seven subsets of A (we don't need to consider $\emptyset$), which are listed here:
$\{1\}$
$\{2\}$
$\{3\}$
$\{1,2\}$
$\{1,3\}$
$\{2,3\}$
$\{1,2,3\}$

For 1 to be an R-*minimal* in B, we must remove $\langle 1,1\rangle$ from R. Similar, arguments apply to $\langle 2,2\rangle$ and $\langle 3,3\rangle$ corresponding to $\{2\}$ and $\{3\}$. Now, for $\{1,2\}$ we must remove $\langle 2,1\rangle$ from R so that 1 is R-*minimal* in $\{1,2\}$, and likewise for $\{1,3\}$ we remove $\langle 3,1\rangle$. Similarly, we remove $\langle 3,2\rangle$ corresponding to $\{2,3\}$. Now, $\{1,2,3\}$ has an R-*minimal* element already, so we are done. Our revised relation R is now:

$$R = \{\langle 1,2\rangle, \langle 1,3\rangle, \langle 2,3\rangle\}$$

This revised relation R well-orders A. This parallels the $<$ (less than) relation in real numbers.

Exercise There is a theorem in Suppes (Theorem 62) as follows: R *well-orders* $A \implies R$ *is asymmetric and transitive in* A. Verify the relation R above satisfies this theorem.

Solution. We first verify that R is asymmetric. We have $\langle 1,2\rangle$ but not $\langle 2,1\rangle$, $\langle 1,3\rangle$ but not $\langle 3,1\rangle$, and lastly $\langle 2,3\rangle$ but not $\langle 3,2\rangle$. Now, we verify transitivity: The only statement that makes sense is $1R2 \wedge 2R3 \implies 1R3$ which is true. Therefore, we have verified that R satisfies the theorem.

Equivalence Relations

We now discuss some important concepts that will appear in some form later, notably in abstract algebra. The reader will benefit by getting comfortable with these topics as early as possible, so we present them here.

- Definition: R *is an equivalence relation* $\iff R$ *is a relation* $\wedge R$ *is reflexive, symmetric, and transitive.*
- Definition: the *R-coset of x* is defined by $R[x] = \{y|xRy\}$.
- Definition: Π *is a partition of* $A \iff \cup\Pi = A \wedge (\forall B)(\forall C)(B \in \Pi \wedge C \in \Pi \wedge B \neq C \implies B \cap C = \emptyset) \wedge (\forall x)(x \in \Pi \implies (\exists y)(y \in x))$.

A *partition* of a set is an exhaustive, mutually exclusive decomposition of a set into subsets, whose union is the set itself. For example, if our set is $A = \{1,2,3\}$, then a partition might be $\Pi = \{\{1\},\{2\},\{3\}\}$ or perhaps $\Pi = \{\{1\},\{2,3\}\}$.

We can see the partitioning through an example. Let us consider a set $A = \{1,2,3,4,5\}$ and construct an equivalence relation, R. Firstly, we know that R must be reflexive, so we provisionally begin with:

$$R = \{\langle 1,1\rangle, \langle 2,2\rangle, \langle 3,3\rangle\ \langle 4,4\rangle\ \langle 5,5\rangle\}$$

We could actually stop here, because R happens to also be symmetric and transitive. However, to make the example slightly more complex, let us add some more members that are symmetric:

$$R = \{\langle 1,1\rangle, \langle 1,2\rangle, \langle 2,1\rangle, \langle 2,2\rangle, \langle 3,3\rangle\ \langle 4,4\rangle\ \langle 5,5\rangle\}$$

where is can be seen that $\langle 1,2\rangle$ and $\langle 2,1\rangle$ have been added.
Now we need to verify that transitivity is satisified:
$1R1 \wedge 1R1 \implies 1R1$
$1R1 \wedge 1R2 \implies 1R2$
$1R2 \wedge 2R1 \implies 1R1$
$1R2 \wedge 2R2 \implies 1R2$
$2R1 \wedge 1R1 \implies 2R1$
$2R1 \wedge 1R2 \implies 2R2$
$2R2 \wedge 2R1 \implies 2R1$

$2R2 \wedge 2R2 \Longrightarrow 2R2$
$3R3 \wedge 3R3 \Longrightarrow 3R3$
$4R4 \wedge 4R4 \Longrightarrow 4R4$
$5R5 \wedge 5R5 \Longrightarrow 5R5$

Now, let us determine the *R-coset of* x for each $x \in A$:
$R[1] = \{1, 2\}$
$R[2] = \{1, 2\}$
$R[3] = \{3\}$
$R[4] = \{4\}$
$R[5] = \{5\}$

Now, we can form a set of all the cosets, which will be a partition of A:
$\Pi = \{\{1, 2\}, \{3\}, \{4\}, \{5\}\}$.

Functions

To help motivate the discussion of *functions*, which presumably many readers will have encountered previously and have a fair degree of experience with, within the set theoretical framework, the following directly quoted passage from Suppes (p.86) may be helpful:

"Since the eighteenth century, clarification and generalization of the concept of a function have attracted much attention. Fourier's representation of 'arbitrary' functions (actually piecewise continuous ones) by trigonometric series encountered much opposition; and later when Weierstrass and Riemann gave examples of continuous functions without derivatives, mathematicians refused to consider them seriously. Even today many textbooks of the differential and integral calculus do not give a mathematically satisfactory definition of functions. An exact and completely general definition is immediate within our set-theoretical framework. A function is simply a many-one relation, that is, a relation which to any element in its domain relates exactly one element in its range."

We now present the formal definition:

Definition: f *is a function* $\Longleftrightarrow$ f *is a relation* $\wedge (\forall x)(\forall y)(\forall z)(xfy \wedge xfz \Longrightarrow y = z)$.

Here is an example from Suppes which we use as an exercise.

Exercise Argue that the following relation is not a function: $f = \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 3, 4\rangle\}$.

Solution:
Appealing to the definition of function, we see that the statement $1f1 \wedge 1f2 \Longrightarrow 1 = 2$ is false, therefore we conclude f is not a function. In other words, the

ordered pairs $\langle 1,1\rangle$ and $\langle 1,2\rangle$ which appear in f have the same first element, 1, but different second elements, 1 and 2, which is not allowed for a function.

Composition of functions is an important concept that appears quite frequently in some of the later topics. The notation for this is $(f \circ g)(x) = f(g(x))$, where f and g are functions, x is an element of a set, and the circle indicates composition. In words, we can think of this as a chain, whereby, firstly the mapping is applied using the the g function, then the result of this is mapped via the f function. Suppes states a formal definition of composition using something called the *relative product*, so for completeness we provide the definitions correspondingly:

Definition: $f \circ g = g/f$.

The relative product is given by the notation g/f and is defined by the following:

Definition: $g/f = \{\langle x,y\rangle \| (\exists z)(xgz \wedge zfy)\}$.

Exercise Given the two functions $f = \{\langle 3,2\rangle, \langle 4,5\rangle, \langle 6,7\rangle\}$ and $g = \{\langle 1,3\rangle, \langle 2,6\rangle\}$, determine the composition $f \circ g$ using the definition g/f.

Solution:
Let us pair up the terms xgz and zfy (in that order), where z is common to both:
$\langle 1,3\rangle, \langle 3,2\rangle$
$\langle 2,6\rangle, \langle 6,7\rangle$

Then we have $g/f = \{\langle 1,2\rangle, \langle 2,7\rangle\}$.

Exercise Prove the following theorem (which is actually part of "Theorem 82" in Suppes): *f and g are functions* $\Longrightarrow$ *$f \circ g$ is a function* .

Solution:
From the definition of a function provided earlier, we need to establish that $h = f \circ g$ is a relation, and also prove that $(\forall x)(\forall y)(\forall z)(xhy \wedge xhz \Longrightarrow y = z)$. For the first part, we appeal to the definition $f \circ g = g/f$ and subsequently note that the definition of g/f is a set of ordered pairs, which is a (binary) relation, also by definition.
For the second part, we need to show: $(\forall x)(\forall y)(\forall z)(xhy \wedge xhz \Longrightarrow y = z)$. Now consider xhy. From the definition of relative product, this means that $(\exists w)(xgw \wedge wfy)$, and likewise for xhz this means that $(\exists w')(xgw' \wedge w'fz)$. Rewriting the hypothesis, we obtain: $(\forall x)(\forall y)(\forall z)((\exists w)(xgw \wedge wfy) \wedge (\exists w')(xgw' \wedge w'fz))$. Now, since g is a function, this implies that $w = w'$. Then, since f is a function and $w = w'$, this implies $y = z$ thereby completing the proof.

We now define the concept of a converse relation, denoted $\breve{R}$, which will be immediately useful in what comes next.

Definition: $\breve{R}= \{\langle x, y\rangle | yRx\}$, where R is a relation.

The following definitions are key:

- Definition: *f is 1-1* $\iff$ *f and $\breve{f}$ are functions* .
- Definition: *f is 1-1* $\implies f^{-1} = \breve{f}$, where f^{-1} is the *inverse* of f.

We note that "1-1" is read "one-to-one." We now illustrate a counter-example to a 1-1 function.

Exercise Show that the following function, $f = \{\langle 1, 2\rangle, \langle 2, 2\rangle, \langle 3, 4\rangle\}$ is not 1-1.
Solution: Let us first determine the converse relation to f: $\breve{f} = \{\langle 2, 1\rangle, \langle 2, 2\rangle, \langle 4, 3\rangle\}$. We see that this relation is not a function because the elements $\langle 2, 1\rangle, \langle 2, 2\rangle$ have the same first entry, but different second entries. Therefore, we conclude f is not 1-1.

We now introduce some additional terminology:

- Definition: Domain of a relation R: $\mathcal{D}R = \{x | (\exists y)(xRy)\}$
- Definition: Range of a relation R: $\mathcal{R}R = \{y | (\exists x)(xRy)\}$. This is also referred to as the *counterdomain* or *converse domain*.
- Definition: *f is a function from set A <u>into</u> set B* $\iff$ *f is a function* $\wedge$ $\mathcal{D}f = A \wedge \mathcal{R}f \subseteq B$.
- Definition: *f is a function from set A <u>onto</u> set B* $\iff$ *f is a function* $\wedge$ $\mathcal{D}f = A \wedge \mathcal{R}f = B$.
- Definition: *Restriction* of the domain of a relation R to a given set A: $R|A = R \cap (A \times \mathcal{R}R)$.
- Definition: the *image* of the set A under R: $R``A = \mathcal{R}(R|A)$.

Exercise What is the image of $A = \{1, 2, 4, 6\}$ under $R = \{\langle 1, 10\rangle, \langle 2, 11\rangle, \langle 5, 15\rangle\}$?

Solution: First, let us determine the restriction $R|A$. We need to obtain the Cartesian product $A \times \mathcal{R}R$ which is:

$$\{\langle 1, 10\rangle, \langle 1, 11\rangle, \langle 1, 15\rangle, \langle 2, 10\rangle, \langle 2, 11\rangle, \langle 2, 15\rangle, \langle 4, 10\rangle, \langle 4, 11\rangle, \langle 4, 15\rangle, \langle 6, 10\rangle, \langle 6, 11\rangle, \langle 6, 15\rangle\}$$

Next, we determine the intersection with R:

$$R|A = R \cap (A \times \mathcal{R}R) = \{\langle 1, 10\rangle, \langle 2, 11\rangle\}$$

Lastly, we obtain the image of A under R:

$$R``A = \mathcal{R}(R|A) = \{10, 11\}$$

It should be kept in mind that we are always dealing with sets when we refer to functions now and later in the material, so if there is any confusion, the reader should always consult the pertinent definitions to see exactly what sets or subsets are being referenced in any manipulations or results. For example, does a particular operation involve the entire set or only some portion of a set. Typically, engineers and scientists may normally think of functions in terms of graphs on a say two-dimensional plot, but in the pure mathematical context, keeping in mind the underlying set framework as has been outlined in the above material is important.

1.4 Peano Axioms of Arithmetic

In the previous section, we built up a basic framework for set theory based on the Zermelo-Frankel axioms. We now establish the foundations of the natural numbers, founded on top of that set theory, by stating the Peano Axioms. Much of this material is based directly on *Schaum's Outline of Theory and Problems of Abstract Algebra* by Frank Ayres and Lloyd R. Jaisingh.

Let there exist a non-empty set $\mathbb{N}$ such that:

- Axiom I: $1 \in \mathbb{N}$.
- Axiom II: For each $n \in \mathbb{N}$ there exists a unique $n^* \in \mathbb{N}$, called the *successor* of n.
- Axiom III: For each $n \in \mathbb{N}$ we have $n^* \neq 1$.
- Axiom IV: If $m, n \in \mathbb{N}$ and $m^* = n^*$, then $m = n$.
- Axiom V: Any subset K of $\mathbb{N}$ having the properties (a) $1 \in K$, (b) $k^* \in K$ whenever $k \in K$ is equal to $\mathbb{N}$.

We now define addition and multiplication on $\mathbb{N}$.

Addition:
(i) $n + 1 = n^*$, for every $n \in \mathbb{N}$
(ii) $n + m^* = (n + m)^*$, whenever $n + m$ is defined.

We then have the following laws for addition. For all $m, n, p \in \mathbb{N}$:

- Closure: $n + m \in \mathbb{N}$
- Commutative: $n + m = m + n$

- Associative: $m + (n + p) = (m + n) + p$
- Cancellation: If $m + p = n + p$, then $m = n$

Exercise Suppose we have two natural numbers, n, m, whereby $n = 1, m = 1^*$. For this restricted case, prove the commutative law for addition: $n + m = m + n$.
Solution:
We have that $n+m = 1+1^*$. Now, from the definition of addition (ii), we obtain $1+1^* = (1+1)^*$. From the definition of addition (i), we obtain $(1+1)^* = (1^*)^*$. Finally, $(1^*)^* = 1^*+1$ from the definition of addition (i) which is equal to $m+n$, thus completing the proof.

Multiplication:
(i) $n \cdot 1 = n$
(ii) $n \cdot m^* = n \cdot m + n$, whenever $n \cdot m$ is defined.

Similar to addition, we have laws for multiplication. For all $m, n, p \in \mathbb{N}$:

- Closure: $n \cdot m \in \mathbb{N}$
- Commutative: $m \cdot n = n \cdot m$
- Associative: $m \cdot (n \cdot p) = (m \cdot n) \cdot p$
- Cancellation: If $m \cdot p = n \cdot p$, then $m = n$.

If we consider both addition and multiplication, there are so-called distributive laws. For all $m, n, p \in \mathbb{N}$:

- D_1: $m \cdot (n + p) = m \cdot n + m \cdot p$
- D_2: $(n + p) \cdot m = n \cdot m + p \cdot m$.

The reader should be cautioned or reminded at this point that the order of the variables and operations in laws such as in the above are important to observe in general. With certain mathematical objects that we will consider, such as *groups*, swapping the positions of variables may not necessarily yield the same results. Therefore, one must carefully consider the definitions relevant to each mathematical object in question.

Exercise How are we justified in asserting that $m \cdot n = n \cdot m$, when $m, n \in \mathbb{N}$?
Solution: The commutative law for multiplication under $\mathbb{N}$ provides the justification.

Exercise Suppose we have $m, n \in \mathbb{N}$ and form the product $p = n \cdot m$. Is the result $p \in \mathbb{N}$?

Solution: Yes. The closure law for multiplication under $\mathbb{N}$ provides the justification.

Note in the above exercises, even though we may have some intuitive sense for the reasons, we must appeal to an exact statement which justifies the assertion. This may be an axiom, definition, theorem, law etc. which has been established for the objects in question. This habit will be useful to form as we consider other contexts in pure mathematics.

1.5 Gödel Theorems

To provide a more complete picture to the reader regarding the underpinnings of mathematics and logic, it is useful to consider the important results by Gödel which are cited in this context. The section is largely adapted from entries in *Stanford Encyclopedia of Philosophy*(https://plato.stanford.edu/entries/goedel/) and *Logic for Mathematicians* by A.G. Hamilton.

There are three theorems that we will consider. The first is called "the Completeness Theorem", the second is "the First Incompleteness Theorem", and the third is "The Second Incompleteness Theorem."

We begin with some definitions adapted from (https://plato.stanford.edu/entries/goedel-incompleteness/):

- *Formal system*: (Roughly) A system of axioms equipped with rules of inference, which allow one to generate new theorems.
- *Complete*: A formal system for which every statement of the system, either the statement or its negation can be derived ("proved") is complete.
- *Consistent*: A formal system for which there is no statement such that the statement and its negation are both derivable is consistent.

The Completeness Theorem

Every valid logical expression is provable. Equivalently, every logical expression is either satisfiable or refutable.

An illustration of how this works is provided in the Stanford reference mentioned earlier, which is reproduced here.
Suppose we have a logical expression $\phi = \forall x_0 \exists x_1 \psi(x_0, x_1)$. We can transform that expression with quantifiers into a list of expressions without quantifiers:

$$\begin{aligned}
\phi_0 &= \psi(x_0, x_1) \\
\phi_1 &= \psi(x_0, x_1) \wedge \psi(x_1, x_2) \\
&\dots \\
\phi_n &= \psi(x_0, x_1) \wedge \ldots \wedge \psi(x_n, x_{n+1})
\end{aligned}$$

Then, there are two cases:
Case 1: For some n, ϕ_n is not satisfiable. This implies the original ϕ is refutable (its negation is provable).
Case 2: Each ϕ_n is satisfiable. This implies the original ϕ is satisfiable.

The Stanford reference states that "logical expression" in the terminology of the theorem is "a well-formed first order formula without identity." Gödel's 1929 dissertation *On the Completeness of the Calculus of Logic* which is the source of the theorem, states the following:
"The main object of the following investigations is the proof of the completeness of the axiom system for what is called the restricted functional calculus, namely the system given in *Whitehead and Russell 1910*, Part I, *1 and *10, and, in a similar way, in *Hilbert and Ackermann 1928* (hereafter cited as H.A.) , III, §5." (Source: Kurt Gödel *Collected Works* Vol.1 by Feferman, et al. (1986) - Page 61.)

The notation and axioms for this system are then specified in the same reference. Briefly, the logic symbols are the same we have seen previously, namely: $\vee, \wedge, \Longrightarrow, \Longleftrightarrow, \neg$ and the axioms are as follows:

- (1) $X \vee X \Longrightarrow X$
- (2) $X \Longrightarrow X \vee Y$
- (3) $X \vee Y \Longrightarrow Y \vee X$
- (4) $(X \Longrightarrow Y) \Longrightarrow (Z \vee X \Longrightarrow Z \vee Y)$
- (5) $(\forall x)F(x) \Longrightarrow F(y)$
- (6) $(\forall x)[X \vee F(x)] \Longrightarrow X \vee (\forall x)F(x)$

The notation $F(x)$ above refers to a "predicate" that can be true or false depending on the value of x. For example, $F(x)$ can be true if $x \in A$, where A is some set.

There are also rules of inference, one of which is the "modus ponens" introduced previously. Namely, $(p \wedge (p \Longrightarrow q)) \Longrightarrow q$ where p and q are statements that can be true or false.

Exercise In the text of Gödel's dissertation, it is stated that $\Longrightarrow$ is used as an abbreviation. Also, it is implied that only $\vee$ and $\neg$ are necessary in order to represent the needed logic. Substantiate this by showing that it is possible to obtain the truth table for $\Longrightarrow$ only using $\vee$ and $\neg$.
Solution:
Recalling the truth table for $\Longrightarrow$:

A	$\Longrightarrow$	B
T	T	T
T	F	F
F	T	T
F	T	F

we can see by inspection that if we negate the A column, and use $\vee$ instead of $\Longrightarrow$ we will reproduce the same truth table. To confirm, let us do this explicity:

$(\neg$	$A)$	$\vee$	B
F	T	T	T
F	T	F	F
T	F	T	T
T	F	T	F

Exercise Is the following logical expression provable (within the above axiom system): $(X \Longrightarrow Y) \vee (X \Longrightarrow Y) \Longrightarrow (X \Longrightarrow Y)$?

Solution: Yes, according to Gödel's Completeness Theorem, since the formula is valid (taking "valid" to mean true for any combination of true or false for the variables X and Y). In fact, in this case it was obtained from Axiom (1) by substituting $X \Longrightarrow Y$ for X, immediately verifying the completeness theorem.

It is worth emphasizing the following point, raised by Gödel in his dissertation. Namely, since the logical axioms are valid and the rules of inference preserve truth, this means that every provable formula (i.e. derived from the axioms using the rules of inference) is valid. Gödel's Completeness Theorem actually states the converse: *Every valid logical expression is provable.* Again, this is within the specific system described above.

The First Incompleteness Theorem

To introduce this section and the next, the following quote from the *Stanford Encyclopedia of Philosophy* entry on Gödel (https://plato.stanford.edu/entries/goedel/) may be illuminating:
"The First Incompleteness Theorem provides a counterexample to completeness by exhibiting an arithmetic statement which is neither provable nor refutable in Peano arithmetic, though true in the standard model. The Second Incompleteness Theorem shows that the consistency of arithmetic cannot be proved

in arithmetic itself."

It is hoped that the previous sections on Peano Axioms and the Completeness Theorem will provide a good starting point for considering these incompleteness theorems.

We now present the First Incompleteness Theorem, based on the entry in the *Stanford Encyclopedia of Philosophy* for Gödel Incompleteness (https://plato.stanford.edu/entries/goedel incompleteness/):

Any consistent formal system F within which a certain amount of elementary arithmetic can be carried out is incomplete; i.e., there are statements of the language of F which can neither be proved nor disproved in F.

At this point, it is useful to introduce the concept of *Gödel numbering* (ref: https://plato.stanford.edu/entries/goedel-incompleteness/sup1.html), which provides a procedure for encoding components of a formal system as natural numbers. The idea is that certain statements can be represented by a number and reasoned about as a number. For example, the statement (Axiom (1) from previous section) $X \vee X \implies X$ might be represented as a natural number like 1329487 and so forth.

To see how Gödel numbers are determined is a unambiguous fashion, we proceed in two steps. Firstly, each symbol of the "language" of the formal system is given a natural number code. For example: the symbol '0' might be assigned code number 1, the plus sign '+' might be 3, the equals sign, '=' might be 4, a variable x_1 might be 13, and $>$ is 14.
The next step is to encode a particular formula, which is a sequence of code numbers, into a final encoded natural number. This is done (apparently following Gödel's original method), by leveraging the prime numbers and the fundamental theorem of arithmetic, which states that every natural number greater than one has a unique prime factorization (up to order of the factors). This means that a given sequence of symbols will have a unique representation in this prime encoding scheme.
Let us represent the primes as $p_1, p_2, p_3, \ldots, p_n, \ldots$. The first few primes in order are: $2, 3, 5, 7, 11$. The next step is to raise each prime to the power of the corresponding symbol code at that position in the symbol sequence. For example, let us take the symbol sequence: $0 + 0 = 0$. The code sequence for this, using our above mapping, is: $1, 3, 1, 4, 1$. To do the final encoding with the primes, we compute $p_1^1 \times p_2^3 \times p_3^1 \times p_4^4 \times p_5^1 = 2^1 \times 3^3 \times 5^1 \times 7^4 \times 11^1 = 2 \times 27 \times 5 \times 2401 \times 11 = 7130970$. So this final result, 7130970, would be the Gödel number for the formula $0 + 0 = 0$.

Exercise Using the above symbol encoding, give an argument for why just representing a given formula by the symbol code sequence is not enough to guarantee a unique Gödel number.

Solution: Consider the formulas $0 + 0 = 0$ vs. $x_1 > 0$. Supposing that both of these are meaningful formulas, we can see that the code sequence for both would be the same, namely: 13141. Therefore, if we left the encoding this way, given the number 13141 we would not be able to determine which of those two formulas was being represented. However, if we do the second step of the process of encoding involving the primes, we would obtain for $x_1 > 0$ the following Gödel number: $2^{13} \times 3^{14} \times 5^1 = 8192 \times 4782969 \times 5 = 195910410240$, which is obviously distinct from 7130970.

Now we have established the concept of Gödel Numbering, we can continue the discussion. In particular, in addition to assigning Gödel numbers to individual formulas, we can imagine assigning a Gödel number to the proof of a formula. If we think of a proof as a sequence of formulas, starting from axioms, applying the inference rules and other logical operations defined in the formal system, we can convert this proof to a Gödel number. For example, taking the *modus ponens* inference rule, suppose we might have something like this $X, X \implies Y, Y, (\neg Y) \implies Z$, where we are trying to prove $(\neg Y) \implies Z$. Suppose that X and $X \implies Y$ are our axioms, then the above sequence could be considered a proof of $(\neg Y) \implies Z$.
As we did before, we can convert each symbol of the eleven symbol sequence to its code number, say: $123, 123, 11, 125, 125, 15, 13, 125, 17, 11, 127$ and then encode to a Gödel number using the primes: $2^{123} \times 3^{123} \times \dots p_{11}^{127}$, where p_{11} is the eleventh prime, 31.

With this in mind, we can then construct the formula which Gödel showed could not be proved or disproved in the particular formal system at hand. Following the Hamilton logic book reference, we define a formula $\mathcal{U}$ in words as follows: for every natural number n, n is not the Gödel number of a proof of the formula $\mathcal{U}$. To clarify, Gödel's theorem states that $\mathcal{U}$ is not a theorem in the formal system, and neither is $\neg(\mathcal{U})$.

Another concept that is useful in this context is *recursive functions*. For those with experience in computer science and software development, the idea of recursive functions may be familiar. According to the Hamilton reference, recursive functions can be constructed from three basic function and three rules which we provide below. Note that the convention used in Hamilton's reference that the set of natural numbers starts with 0 as opposed to 1, and the symbol D_N is used to represent this set of natural numbers starting from 0. The symbol D_N^k then represents a k-dimensional cartesian product set of natural numbers. For example, $D_N^2 = \{\langle 0,0\rangle, \langle 1,0\rangle, \langle 2,0\rangle, \dots\}$, and then $\langle 2,3\rangle$ and $\langle 5,11\rangle$ are two particular elements of that set.

The basic functions are:

- 1. The zero function $z : D_N \to D_N$, given by $z(n) = 0$ for every $n \in D_N$. The notation $z : D_N \to D_N$ is read as "z is the function which maps elements in D_N to D_N". Note that the arrow is not the same and the logical implication $\Longrightarrow$ previously discussed. So in this case, the function z maps (i.e. connects) every natural number in D_N to the natural number 0.

- 2. The successor function $s : D_N \rightarrow D_N$, given by $s(n) = n + 1$ for every $n \in D_N$.

- 3. The projection functions $p_i^k : D_N^k \rightarrow D_N$, given by $p_i^k(n_1, \ldots, n_k) = n_i$, for every $n_1, \ldots, n_k \in D_N$.

The rules are as follows:

- R1. Composition. If $g : D_N^j \rightarrow D_N$ and $h_i : D_N^k \rightarrow D_N$ for $1 \leq i \leq j$, then $f : D_N^k \rightarrow D_N$, defined by $f(n_1, \ldots, n_k) = g(h_1(n_1, \ldots, n_k), \ldots, h_j(n_1, \ldots, n_k))$ is obtained by composition from g and $h_1, \ldots, h_j$.

- R2. Recursion. If $g : D_N^k \rightarrow D_N$ and $h : D_N^{k+2} \rightarrow D_N$, then the function $f : D_N^{k+1} \rightarrow D_N$, defined by $f(n_1, \ldots, n_k, 0) = g(n_1, \ldots, n_k)$, and $f(n_1, \ldots, n_k, n + 1) = h(n_1, \ldots, n_k, n, f(n_1, \ldots, n_k, n))$ is said to be obtained by recursion from g and h. We can also have recursion with all the n_i absent, as in $f(0) = a$ and $f(n + 1) = h(n, f(n))$, where a is a fixed member of D_N.

- R3. Least Number Operator. Let $g : D_N^{k+1} \rightarrow D_N$ be any function with the property that for each $n_1, \ldots, n_k \in D_N$ there is at least one $n \in D_N$ such that $g(n_1, \ldots, n_k, n) = 0$. Then the function $f : D_N^k \rightarrow D_N$, defined by $f(n_1, \ldots, n_k)$ = least number $n \in D_N$ such that $g(n_1, \ldots, n_k, n) = 0$, is said to be obtained from g by the use of the least number operator.

Let us consider some examples to illustrate the above functions and rules.

Taking the projection function p_2^3, we can apply it to $\langle 4, 6, 9 \rangle$ and obtain $p_2^3(4, 6, 9) = 6$. This is the second value in the triplet $\langle 4, 6, 9 \rangle$ extracted (or "projected").

Now, moving to composition, take $g(n_1, n_2, n_3) = n_1 n_2 + n_3$ and $h_1(m_1, m_2) = m_1 + m_2, h_2(m_1, m_2) = m_1 m_2, h_3(m_1, m_2) = m_1 + 2m_2$. Then, the composition is $f(m_1, m_2) = g(h_1(m_1, m_2), h_2(m_1, m_2), h_3(m_1, m_2)) = (m_1 + m_2)(m_1 m_2) + (m_1 + 2m_2)$.

Lastly, let us consider recursion. Let $h(m_1, m_2) = m_1 + m_2$ and $f(0) = 1$. Then, $f(1) = h(0, f(0)) = 0 + 1 = 1$, $f(2) = h(1, f(1)) = 1 + 1 = 2$, $f(3) = h(2, f(2)) = 2 + 2 = 4$, $f(4) = h(3, f(3)) = 3 + 4 = 7$, and so forth.

We now show that the "addition function", which is one of the basic operators in our formal system for arithmetic, is recursive. In other words, it can be constructed from the above basic functions and finite application of the rules.

We show how this example (taken from Hamilton) is done.

We want to show that the addition function $f_1(p,q) = p + q$ can be constructed as recursive. If we take the projection functions p_1^1, p_3^3, and the successor function s we can form the following:

$$\begin{aligned} f_1(m,0) &= p_1^1(m) \\ f_1(m,n+1) &= s(p_3^3(m,n,f_1(m,n)) \end{aligned}$$

Let's check this for $f_1(2,3)$ which we anticipate should be $2+3=5$. We start by working backwards from the final function:

$$\begin{aligned} f_1(2,3) &= s(p_3^3(2,2,f_1(2,2)) \\ f_1(2,2) &= s(p_3^3(2,1,f_1(2,1)) \\ f_1(2,1) &= s(p_3^3(2,0,f_1(2,0)) \\ f_1(2,0) &= p_1^1(2) = 2 \end{aligned}$$

We have reached the "base case" and now we can evaluate the function working in the forward direction:

$$\begin{aligned} f_1(2,0) &= p_1^1(2) = 2 \\ f_1(2,1) &= s(p_3^3(2,0,f_1(2,0)) = s(p_3^3(2,0,2)) = 3 \\ f_1(2,2) &= s(p_3^3(2,1,f_1(2,1)) = s(p_3^3(2,1,3)) = 4 \\ f_1(2,3) &= s(p_3^3(2,2,f_1(2,2)) = s(p_3^3(2,2,4)) = 5 \end{aligned}$$

Exercise Show how to construct the multiplication function, $f_2(m,n) = mn$ as a recursive, using the addition function, f_1 defined above recursively, as well as the appropriate projection functions.

Solution If we consider what mn means as an addition problem, this can give us a clue how to proceed. Namely, we know that $mn = \overbrace{m + m + \ldots + m}^{n}$. So, if we start with a base case of 0, and then add m to that base value n times, we should get the desired result. Putting this into the recursive format, and leveraging the addition function, f_1 defined earlier, we obtain:

$$\begin{aligned} f_2(m,0) &= z(m) = 0, \text{ where } z \text{ is the zero function defined earlier} \\ f_2(m,n+1) &= f_1(p_3^3(m,n,f_2(m,n)), p_1^3(m,n,f_2(m,n))) \end{aligned}$$

Let's check this for a simple case to see if it works. Take $f_2(3,5)$ which we expect should be 15. As before let us start by working backwards from the final function:

$$
\begin{aligned}
f_2(3,5) &= f_1(f_2(3,4),3) \\
f_2(3,4) &= f_1(f_2(3,3),3) \\
f_2(3,3) &= f_1(f_2(3,2),3) \\
f_2(3,2) &= f_1(f_2(3,1),3) \\
f_2(3,1) &= f_1(f_2(3,0),3) \\
f_2(3,0) &= 0
\end{aligned}
$$

We have reached the "base case", so now we can evaluate the function working in the forward direction:

$$
\begin{aligned}
f_2(3,0) &= 0 \\
f_2(3,1) &= f_1(f_2(3,0),3) = 0+3 = 3 \\
f_2(3,2) &= f_1(f_2(3,1),3) = 3+3 = 6 \\
f_2(3,3) &= f_1(f_2(3,2),3) = 6+3 = 9 \\
f_2(3,4) &= f_1(f_2(3,3),3) = 9+3 = 12 \\
f_2(3,5) &= f_1(f_2(3,4),3) = 12+3 = 15
\end{aligned}
$$

We now expand the idea of recursion to apply to relations more generally. First we define a characteristic function (adapted from Hamilton Def 6.17):

Definition Let R be a k-place relation on D_N. The *characteristic function* of R, denoted C_R is defined by

$$
C_R(n_1,\ldots,n_k) = \begin{cases} 0 \text{ if } R(n_1,\ldots,n_k) \\ 1 \text{ if } \neg R(n_1,\ldots,n_k) \end{cases}
$$

Note that previously we only considered binary relations in detail (i.e. two place), but we can extend that to three place, four place, etc. For example, we can think of an entry in a three place relations using the ordered pair notation as $\langle\langle x,y\rangle,z\rangle$ and describe it generally as the set $R(x,y,z)$. If a particular triplet, in this case, is present in the set $R(x,y,z)$, then the relation is true otherwise it is false.

Given the definition of a characteristic function of a relation, we make the following definition (Hamilton Def. 6.18):

Definition A relation R on D_N is *recursive* if its characteristic function is a recursive function.

We adapt an example from Hamilton to illustrate this. Define a binary relation $R(m,n)$ on D_N to be true whenever $m+n$ is odd. If we can show that the associated characteristic function is recursive, then by definition, so is the relation R. The characteristic function is given as:

$$C_R(m,n) = \begin{cases} 0 \text{ if } R(m,n) \text{ is true, which means } m+n \text{ is odd} \\ 1 \text{ if } \neg R(m,n) \text{ is true, which means } m+n \text{ is even} \end{cases}$$

In order to proceed, we assume a recursive function exists which computes the remainder after division by 2. Call this function $rm(n)$. For example, $rm(5) = 1$ and $rm(8) = 0$. In other words, this function returns 1 for odd arguments and 0 for even. This function rm is actually shown to be recursive in Hamilton, so we leverage that result. We know that the addition function $f_2(m,n)$ is recursive, since we showed that earlier. We almost have all the necessary building blocks. We will need a recursive function to represent a constant. It is also stated in Hamilton that such a function $f(n) = k$, where $k \in D_N$ is some constant, is indeed recursive.

However, we still need a recursive function which returns 0 when the argument is non-zero and returns 1 when the argument is 0, so that we can construct our characteristic function $C_R(m,n)$ to have the correct result for when $R(m,n)$ is true (i.e. when $m+n$ is odd). We now construct such a recursive function, and call it $\overline{sg}$, as is done in Hamilton. We now define it as follows:

$$\begin{aligned} \overline{sg}(0) &= f(0), \text{ where } f(n) = 1, \text{ the constant function } 1 \\ \overline{sg}(n+1) &= z(p_1^2(n, \overline{sg}(n)) = 0 \end{aligned}$$

Given all these functions, we can now construct the characteristic function by using "composition" (i.e. Rule R1.):

$$C_R(m,n) = \overline{sg}(rm(f_2(m,n)))$$

Since each individual function is recursive (i.e. constructed using the base functions and rules R1-R3), and we constructed C_R using the composition rule, the resulting function C_R is also recursive. This then implies that the relation $R(m,n)$ is also recursive (by the earlier definition) and we are done.

We note that the definition of recursiveness applies to sets of numbers. In particular, if $A \subseteq D_N$, the set A is recursive if the characteristic function of the relation $\in A$ is recursive. An example is given to show that the set of even numbers is recursive. The remainder function rm discussed above, which is recursive, can act as the characteristic function in this case. In particular, $n \in A$, where A is the set of even natural numbers, then $rm(n) = 0$ otherwise $rm(n) = 1$.

An important connection exist between recursive relations and the formal system we are considering, namely the system of first order arithmetic. In particular, a recursive relation is “expressible” in this system, which is referred to as $\mathcal{N}$.

To continue this discussion towards a proof of Gödel's First Incompleteness Theorem, let us consider the following recursive relation $W(m,n)$ defined on D_N, (the natural numbers), which is expressible in $\mathcal{N}$ (adapted from Proposition 6.29 in Hamilton)

$W(m,n)$ holds if and only if m is the Gödel number of a *wf* $\mathcal{A}(x_1)$ in which x_1 occurs free, and n is the Gödel number of a proof in $\mathcal{N}$ of $\mathcal{A}(0^{(m)})$.

Then, since W is expressible in $\mathcal{N}$, this means there is a well-formed formula (*wf*) $\mathcal{W}(x_1, x_2)$ such that:

If $W(m,n)$ holds then $\vdash_{\mathcal{N}} \mathcal{W}(0^{(m)}, 0^{(n)})$, where $\vdash_{\mathcal{N}}$ means that the formula is derivable or is a theorem of the system, $\mathcal{N}$. The symbol $0^{(m)}$ corresponds to the natural number m. Conversely, if $W(m,n)$ does not hold then $\vdash_{\mathcal{N}} \neg\mathcal{W}(0^{(m)}, 0^{(n)})$.

Let us examine expressibility with an example adapted from Hamilton (Example 6.8). The function $f : D_N \to D_N$ given by $f(m) = 2m$ is expressible in $\mathcal{N}$. Let $\mathcal{A}(x_1, x_2)$ be the *wf* $x_2 = x_1 \times 0^{(2)}$. By using the axioms of $\mathcal{N}$ (which is done in Hamilton) the two following required statements can be established.

$$\begin{aligned}
&\text{if } n = 2m \text{ then } \vdash_{\mathcal{N}} 0^{(n)} = 0^{(m)} \times 0^{(2)} \\
&\text{if } n \neq 2m \text{ then } \vdash_{\mathcal{N}} \neg(0^{(n)} = 0^{(m)} \times 0^{(2)})
\end{aligned}$$

To illustrate the first couple of steps in the proof, we have:

$$\begin{aligned}
0^{(m)} \times 0^{(2)} &= 0^{(m)} \times 0'' && \text{(Reason: notation)} \\
&= (0^{(m)} \times 0') + 0^{(m)} && \text{(Reason: Axiom N6*)} \\
&\dots \\
&= 0^{(n)}
\end{aligned}$$

Axiom N6* is given in Hamilton as: $(\forall x_1)(\forall x_2)(x_1 \times x_2' = (x_1 \times x_2) + x_1)$. We note that the prime on x_2 means the “successor” of x_2. For example, 3 is the successor of 2 in the natural numbers.

We now state Gödel's First Incompleteness Theorem in precise language, followed by the proof, based on Hamilton (Proposition 6.32):

Under the assumption that $\mathcal{N}$ is ω-consistent, the wf $\mathcal{U}$ is not a theorem of $\mathcal{N}$, nor is its negation. Hence, if $\mathcal{N}$ is ω-consistent, then $\mathcal{N}$ is not complete.

The well-formed formula $\mathcal{U}$ is defined as $(\forall x_2)\neg\mathcal{W}(0^{(p)}, x_2)$, where p is the Gödel number of the *wf* $(\forall x_2)\neg\mathcal{W}(x_1, x_2)$.

The term ω-consistent asserts that if each well-formed formula $\mathcal{A}(0^{(n)})$ is a theorem of $\mathcal{N}$ (where $n \in D_N$), then $\neg(\forall x_1)\mathcal{A}(x_1)$ is not a theorem. We note that in Hamilton, Remark 6.33 states that "We have explicitly stated the assumption of ω-consistency, although there is an obvious demonstration that $\mathcal{N}$ is ω-consistent, using the model N (N refers to the an interpretation of arithmetic in natural numbers, including the distinguished element 0, addition and multiplication, the successor function, and the equality relation $=$).

The proof is below:

Part I: We show that $\mathcal{U}$ is not a theorem of $\mathcal{N}$.
Suppose that $\mathcal{U}$ is a theorem of $\mathcal{N}$. Let q be the Gödel number of a proof of $\mathcal{U}$ in $\mathcal{N}$. Let p be the Gödel number of $(\forall x_2)\neg\mathcal{W}(x_1, x_2)$. Thus $W(p, q)$ holds. W is expressible in $\mathcal{N}$ so we have $\vdash_{\mathcal{N}} \mathcal{W}(0^{(p)}, 0^{(q)})$. In other words, $\mathcal{W}$ is derivable from the axioms of $\mathcal{N}$.
But we assumed $\mathcal{U}$ is theorem, which means $\vdash_{\mathcal{N}} (\forall x_2)\neg\mathcal{W}(0^{(p)}, x_2)$, and so $\vdash_{\mathcal{N}} \neg\mathcal{W}(0^{(p)}, 0^{(q)})$, using Axiom K5 and modus ponens. This contradicts the consistency of $\mathcal{N}$ (note that ω-consistency implies consistency), so $\mathcal{U}$ cannot be a theorem of $\mathcal{N}$.

Axiom K5 of the system is: $((\forall x_i)\mathcal{A}(x_i) \Longrightarrow \mathcal{A}(t))$.

Part II: We show that $\neg\mathcal{U}$ is not a theorem of $\mathcal{N}$.
From Part I, we know that $\mathcal{U}$ is not a theorem of $\mathcal{N}$, i.e. there is no proof in $\mathcal{N}$ of $\mathcal{U}$, so for no number q is q the Gödel number of a proof in $\mathcal{N}$ of $\mathcal{U}$, i.e. of $(\forall x_2)\neg\mathcal{W}(0^{(p)}, x_2)$, thus $W(p, q)$ does not hold for any number q. Therefore

$$\vdash_N \neg\mathcal{W}(0^{(p)}, 0^{(q)}) \text{ for every } q.$$

By ω-consistency, then,

$$\neg(\forall x_2)\neg\mathcal{W}(0^{(p)}, x_2)$$

is not a theorem of $\mathcal{N}$ i.e. $(\neg\mathcal{U})$ is not a theorem of $\mathcal{N}$.
End of proof.

Since we have shown that a formula exists in $\mathcal{N}$, called $\mathcal{U}$ which is not a theorem, and the negation $(\neg\mathcal{U})$ is not a theorem, this means that $\mathcal{N}$ is not complete.

We now consider some exercises to better understand the proof.

Exercise In Part I of the proof, it was stated that there was a contradiction to the consistency of $\mathcal{N}$. Please explain why this statement could be made.

Solution First recall the definition of consistency: *A formal system for which there is no statement such that the statement and its negation are both derivable is consistent.* In Part I of the proof, under the assumption that $\mathcal{U}$ was a theorem of $\mathcal{N}$, we obtained $\vdash_{\mathcal{N}} \mathcal{W}(0^{(p)}, 0^{(q)})$ and $\vdash_{\mathcal{N}} \neg\mathcal{W}(0^{(p)}, 0^{(q)})$. Since we know that $\mathcal{N}$ is consistent, obtaining both these results, as one is the negation of the other, violates consistency. From this contradiction, we conclude that our original assumption that $\mathcal{U}$ is a theorem of $\mathcal{N}$ must be false.

Exercise In Part II of the proof, please explain how ω-consistency implies that if we have $\vdash_N \neg\mathcal{W}(0^{(p)}, 0^{(q)})$ for every q, we get that $\neg(\forall x_2)\neg\mathcal{W}(0^{(p)}, x_2)$ is not a theorem of $\mathcal{N}$.

Solution Recalling that the term ω-consistent asserts that if each well-formed formula $\mathcal{A}(0^{(q)})$ is a theorem of $\mathcal{N}$ (where $q \in D_N$), then $\neg(\forall x_1)\mathcal{A}(x_1)$ is not a theorem, we first let $\mathcal{A}(0^{(q)}) = \neg\mathcal{W}(0^{(p)}, 0^{(q)})$. (Note that we substituted q for n in the original statement of ω-consistency). Then, we have that $\neg(\forall x_2)\mathcal{A}(x_2)$ is not a theorem. (Note that we have substituted x_2 for x_1). Substituting in for $\mathcal{A}$, we obtain finally that $\neg(\forall x_2)\neg\mathcal{W}(0^{(p)}, x_2)$ is not a theorem of $\mathcal{N}$.

The Second Incompleteness Theorem

We now present Gödel Second Incompleteness Theorem, largely following the Stanford Encyclopedia of Philosophy entry on Kurt Gödel (https://plato.stanford.edu/entries/goedel/). We note that in this reference, the formal system is called P and "contains" Peano arithmetic. (Note: Please be aware that in this particular case of Peano arithmetic, the minimal value is 0 as opposed to 1).

In summary, the Second Incompleteness Theorem establishes that the consistency of P cannot be *proved* in P. Here, "consistency' is meant in the technical term we defined earlier, namely in a consistent formal system, you cannot derive both a statement and its negation. We state the theorem below in precise terms:

If P is consistent, then $\neg Prov(\ulcorner \underline{0 = 1} \urcorner)$ is not provable from P.

In the above statement of the theorem, the relation $Prov(y)$ is true whenever the "sentence" (formula) with Gödel number y is provable in the system P. The sequence $\ulcorner x \urcorner$ means the Gödel number of the sentence x. The underline signifies the formal term corresponding to the natural number code of the argument. For example, assume the Gödel number of the statement $0 = 1$ is 57, i.e. $\ulcorner 0 = 1 \urcorner = 57$. Then, $\underline{57}$ would be how that number is represented in the formal system P. In particular, it would be whatever formula represents 57 applications of the successor function on 0, say $0^{(57)}$. It should be emphasized that the sentence $\neg Prov(\ulcorner \underline{0 = 1} \urcorner)$ is a statement which asserts the consistency of P.

In order to give the proof of the Second Incompleteness Theorem, we define a "sentence" ϕ in the following manner:

$$P \vdash (\phi \Longleftrightarrow \neg Prov(\ulcorner \underline{\phi} \urcorner))$$

In words, the ϕ in the above definition is not provable in P. It is shown in the Stanford reference that $P \nvdash \phi$, meaning ϕ cannot be derived in P, (i.e. is not a theorem in P.)

We now continue with the proof of the Second Incompleteness Theorem (again following the Stanford reference):

Utilize the definition of ϕ stated above. It is first assumed that the reasoning to obtain the statement "if $P \vdash \phi$, then $P \vdash 0 \neq 1$" can be stated in P. This yields: $P \vdash (Prov(\ulcorner \underline{\phi} \urcorner) \Longrightarrow Prov(\ulcorner \underline{0 = 1} \urcorner))$. Then we also have, by the definition of ϕ, $P \vdash (\neg Prov(\ulcorner \underline{0 = 1} \urcorner) \Longrightarrow \phi)$. Since we know that actually $P \nvdash \phi$, this means that $P \nvdash \neg Prov(\ulcorner \underline{0 = 1} \urcorner)$. This means that a statement which asserts the consistency of P cannot be proved in P.

End of proof.

Exercise It was stated in the proof from the definition of ϕ we can state $P \vdash (\neg Prov(\ulcorner \underline{0 = 1} \urcorner) \Longrightarrow \phi)$. Please explain how this is possible using logical manipulations.

Solution We can rewrite $P \vdash (\phi \Longleftrightarrow \neg Prov(\ulcorner \underline{\phi} \urcorner))$ as $P \vdash ((\phi \Longrightarrow \neg Prov(\ulcorner \underline{\phi} \urcorner)) \wedge (\neg Prov(\ulcorner \underline{\phi} \urcorner) \Longrightarrow \phi))$ and as both terms must be true for the $\wedge$ to be true, the second term corresponds to part of what we seek. In the proof, we also have $P \vdash (Prov(\ulcorner \underline{\phi} \urcorner) \Longrightarrow Prov(\ulcorner \underline{0 = 1} \urcorner))$. Writing this expression in the contrapositive, we obtain $P \vdash (\neg Prov(\ulcorner \underline{0 = 1} \urcorner) \Longrightarrow \neg Prov(\ulcorner \underline{\phi} \urcorner))$. So, in summary, we have

$$P \vdash (\neg Prov(\ulcorner \underline{0 = 1} \urcorner) \Longrightarrow \neg Prov(\ulcorner \underline{\phi} \urcorner))$$
$$P \vdash (\neg Prov(\ulcorner \underline{\phi} \urcorner) \Longrightarrow \phi)$$

and thus we can finally write

$$P \vdash ((\neg Prov(\ulcorner \underline{0 = 1} \urcorner) \Longrightarrow \phi).$$

Exercise It was emphasized that the sentence $\neg Prov(\ulcorner \underline{0 = 1} \urcorner)$ is a statement which asserts the consistency of P. Does this make sense, since it is only a single formula and the definition of consistency is a statement about a formula and the negation of that formula?

Solution This does makes sense, if we take consider the negation of $0 = 1$, which would be $0 \neq 1$. In the system P, which contains Peano arithmetic, there is an axiom (Axiom III in our earlier section on Peano Arithmetic) which states: "Axiom III: For each $n \in \mathbb{N}$ we have $n^* \neq 1$." If we replace 1 with 0 (since the system P we are using for the Second Incompleteness Theorem uses 0 as the minimal element), we see that n^*, the successor of any n, does not equal the minimal element. Thus, $0 \neq 1$ is true by Axiom III and indeed derivable in one step, if we let $n = 0$ and $n^* = 1$. Therefore, if P is consistent, then we must have $\neg Prov(\ulcorner \underline{0 = 1} \urcorner)$ since $Prov(\ulcorner \underline{0 \neq 1} \urcorner)$ as we just showed. However, note carefully that we cannot actual *prove* $\neg Prov(\ulcorner \underline{0 = 1} \urcorner)$ in P, as we learned by the Second Incompleteness Theorem.

Chapter 2

Methods of Proof

In this chapter, we will discuss various methods of proof and provide illustrations of each. In the first chapter, we saw some examples of proofs without dwelling on the process of doing the proof itself. The emphasis in this book is to gain experience with reading and writing proofs within various topics in pure mathematics. It is important to both read *and* write proofs in order to become comfortable with them. The exit goal of this chapter is that the reader will have a general knowledge of the various types of proofs and be able to read and write such proofs.

One may see proofs that are written as lists, say with the statement on the left and the justification on the right, or written more like prose, where more usual written language is combined with the mathematics and logic. In the previous chapter, there were examples of each type.

Much of the material in this chapter is based on the YouTube video series "Math Major Basics" by MathDoctorBob (https://www.youtube.com/playlist?list=PLF2DF6C3C8015DF5F).

We first remark that a *theorem* is a set of valid statements chained together by logical "ands" (i.e. $\wedge$). If all the statements are true and the conclusion is true, then the theorem is *proved*.

2.1 Induction

In the *induction* type of proof, the statements in question will be associated with an index which is used to construct the proof.

2.1.1 Weak Induction

Let us begin by stating the "principle of mathematic induction", based on the presentation in Schaum's Outline of Theory and Problems of Set Theory and

Related Topics (2nd Ed), by Seymour Lipschutz (SST).

Let $A(n)$ be an assertion defined on the set of positive integers, $\mathbb{P}$. $A(n)$ is true or false for each integer $n \geq 1$. Suppose $A(n)$ has the following two properties:

- $A(1)$ is true (sometimes called the "base case")
- $A(n+1)$ is true whenever $A(n)$ is true (sometimes called the "inductive hypothesis")

Then $A(n)$ is true for every $n \geq 1$.

Let us see an example of this in practice.

We wish to prove that $\Sigma_{i=1}^{n} i = \frac{n(n+1)}{2}$ using (weak) induction. That is, the sum of the first n positive integers is equal to $\frac{n(n+1)}{2}$. For example: $1+2+3+4+5 = 15, \frac{5(6)}{2} = \frac{30}{2} = 15\checkmark$.

We can first check the base case: $\Sigma_{i=1}^{1} i = 1, \frac{1(1+1)}{2} = 1\checkmark$. Thus $A(1)$ is true, where $A(n)$ is the assertion that $\Sigma_{i=1}^{n} i = \frac{n(n+1)}{2}$.

Now, suppose that $A(n)$ is true. We then check $A(n+1)$. This yields: $\Sigma_{i=1}^{n+1} i = \frac{(n+1)(n+2)}{2}$. If $A(n+1)$ is true, then $A(n+1) = A(n) + (n+1)$. So we check:

$$
\begin{aligned}
\frac{(n+1)(n+2)}{2} &= \frac{n(n+1)}{2} + (n+1)? \\
\frac{n(n+1)}{2} + (n+1) &= \frac{n(n+1) + 2(n+1)}{2} \\
\frac{n(n+1) + 2(n+1)}{2} & \\
&= \frac{(n+1)(n+1)}{2}\checkmark
\end{aligned}
$$

Thus, we conclude that $A(n)$ is true for $n \geq 1$ which completes our proof.

We consider a second example, taken from Schaum's Outline of Theory and Problems of Abstract Algebra (2nd Ed.) by Frank Ayres, Jr. and Lloyd R. Jaisingh (SAA). This is associated with the Peano Postulates of Arithmetic that we considered in Chapter 1. We wish to prove the Closure Law for Addition, namely $n + m \in \mathbb{N}$, for all $n, m \in \mathbb{N}$, using induction. Recall that $\mathbb{N}$ is the set of natural numbers (which starts at 1 in this case). We proceed as follows:

We use the two definitions associated with addition on $\mathbb{N}$, namely (i) $n+1 = n^*$, for every $n \in \mathbb{N}$, where n^* is the successor of n, and (ii) $n+m^* = (n+m)^*$, whenever $n+m$ is defined (i.e. whenever $n+m$ is a natural number), as well as Peano Postulate II: For each $n \in \mathbb{N}$, there exists a unique $n^* \in \mathbb{N}$, called the successor of n.

We first fix n as some natural number, and define the proposition $P(m)$: $n+m \in \mathbb{N}$ for every $m \in \mathbb{N}$. That is, $P(m)$ is true, whenever $n+m$ is a natural number. First, let us determine if $P(1)$ is true (i.e. the base case). Since $n+1 = n^*$, from Postulate II we know that $n^* \in \mathbb{N}$, so $P(1)$ is true. Now, for the inductive hypothesis portion. Suppose that $P(m)$ is true, then we want to see if $P(m+1)$ is true, which is the same as $P(m^*)$ by Definition (i). Now, $P(m^*)$ is true if $n+m^* \in \mathbb{N}$. From Definition (ii), we have that $n+m^* = (n+m)^*$ whenever, $n+m$ is defined (i.e. is a natural number). Since we assumed $P(m)$ was true, this means that $n+m$ is a natural number, and by Postulate II, $(n+m)^*$ is also a natural number, so we are done.

Exercise (Adapted from SST) Prove the following assertion (for $n \geq 1$) using (weak) induction:

$$A(n) : 1 + 2 + 2^2 + \ldots + 2^{(n-1)} = 2^n - 1$$

Solution We begin by checking if the base case is true, namely $A(1)$. We obtain $2^{(1-1)} = 1 = 2^1 - 1\checkmark$. Now we check the induction hypothesis portion. Suppose $A(n)$ is true, then if $A(n+1) = A(n) + 2^{(n+1-1)} = A(n) + 2^n = 2^n - 1 + 2^n = 2(2^n) - 1 = 2^{(n+1)} - 1$, we are done. Since $A(n+1) = 2^{(n+1)} - 1$ we see that the proof is therefore complete.

2.1.2 Strong Induction

There is a second formulation of induction called "complete" or "strong" induction. It is defined in the SST reference as follows:

Let $A(n)$ be an assertion defined on the set of positive integers, $\mathbb{P}$. Suppose $A(n)$ has the following two properties:

- $A(1)$ is true.
- $A(n)$ is true whenever $A(k)$ is true for $1 \leq k < n$.

Then $A(n)$ is true for every $n \geq 1$.

The process for doing strong/complete induction proofs is essentially identical to the weak induction. The interested reader can see the MathDoctorBob videos cited above for some examples. We briefly outline one such example.

In order to prove that the following formula, called "Binet's Formula" produces the n'th Fibonacci number, it is argued in the MathDoctorBob videos (video BM4.1) that strong induction can be used. Recall that Fibonacci numbers, F_n, are obtained by taking the sum of the previous two Fibonacci numbers beginning with $F_1 = 1$ and $F_2 = 1$, whereby $F_3 = F_2 + F_1 = 1 + 1 = 2$, and $F_4 = F_3 + F_2 = 2 + 1 = 3$ and so forth. Binet's formula gives the n'th Fibonacci number as:

$$F_n = \frac{1}{\sqrt{5}}\left[\phi_1^n - \phi_2^n\right]$$

where

$$\phi_1 = \frac{1+\sqrt{5}}{2}, \phi_2 = \frac{1-\sqrt{5}}{2}.$$

In order to proceed with the proof, it is necessary to assume (in the inductive hypothesis step) that *two* immediately prior values of the Binet formula are true, since we stated that $F_n = F_{n-1} + F_{n-2}$. Thus, we must assume that both F_{n-1} and F_{n-2} are true and then inquire about F_n. In the weak induction method, we only had to assume that one immediate prior value is true for the assertion in question.

To sketch the inductive hypothesis step, we would assume that F_{n-1} and F_{n-2} are both true (i.e. Binet's formula holds for both of those cases) and then check to see if Binet's formula for F_n is then true:

$$F_n = \frac{1}{\sqrt{5}}\left[\phi_1^n - \phi_2^n\right] = \frac{1}{\sqrt{5}}\left[\phi_1^{n-1} - \phi_2^{n-1}\right] + \frac{1}{\sqrt{5}}\left[\phi_1^{n-2} - \phi_2^{n-2}\right]?$$

We will examine a portion of the proof in the following exercise.

Exercise In order to establish part of the proof above, we need to determine if $\phi_1^n = \phi_1^{(n-1)} + \phi_1^{(n-2)}$ (The factors of $\frac{1}{\sqrt{5}}$ are irrelevant here). Please show this equation is indeed correct.

Solution We proceed with some algebraic manipulation:

$$\begin{aligned}
\phi_1^n &= \phi_1^{(n-1)} + \phi_1^{(n-2)}? \\
\phi_1^n - \phi_1^{(n-1)} &= \phi_1^{(n-2)}? \\
\phi_1^n(1 - \phi_1^{-1}) &= \phi_1^n \phi_1^{-2}? \\
1 - \phi_1^{-1} &= \phi_1^{-2}?
\end{aligned}$$

Now, $1-\phi_1^{-1} = 1-\frac{2}{1+\sqrt{5}} = \frac{1+\sqrt{5}-2}{1+\sqrt{5}} = \frac{\sqrt{5}-1}{\sqrt{5}+1} = \frac{\sqrt{5}-1}{\sqrt{5}+1}\frac{\sqrt{5}+1}{\sqrt{5}+1} = \frac{4}{\left(\sqrt{5}+1\right)^2} = \phi_1^{-2}\checkmark$. We would proceed in a similar way to verify the other part of the assertion involving ϕ_2 to complete the inductive hypothesis portion of the proof.

Let us verify the base case in a second exercise.

Exercise Show that the base case $F_1 = \frac{1}{\sqrt{5}}(\phi_1 - \phi_2) = 1$, i.e. the first Fibonacci number.
Solution Substituting for ϕ_1 and ϕ_2 we obtain $F_1 = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right) - \left(\frac{1-\sqrt{5}}{2}\right)\right) = \frac{1}{\sqrt{5}}\frac{2\sqrt{5}}{2} = 1$.

2.2 Proof by Contradiction

Proof by contradiction is a very useful method of proof which is used quite frequently. The usual pattern is one assumes the converse of what is being asserted, and then following a sequence of logical deductions arriving at a contradiction. Once this contradiction has been uncovered, then it implies the original assumption must have been false thus completing the proof. Here is an example to illustrate the points.

Prove that there is no rational number whose square is 12. (Note: This was originally an exercise in Rudin PMA 3rd Edition).

Proof:

Suppose that there was a rational number whose square was 12. This means that

$$\left(\frac{p}{q}\right)^2 = 12$$

for integers p, q. We assume that p and q are not both even. In other words, all common factors of 2 have been cancelled. This can always been done for a rational number.

Upon expansion we obtain:

$$p^2 = 12q^2 = 2\left(6q^2\right)$$

This implies that p^2 and p are both even. Thus q is odd. So we write $p = 2m$ for integer m and obtain:

$$(2m)^2 = 4m^2 = 12q^2$$

This implies that

$$m^2 = 3q^2$$

Since q is assumed to be odd, q^2 is odd, and $3q^2$ is also odd. Now, m^2 must be therefore be odd, and likewise for m. Therefore, for integers r, n we can rewrite the above as:

$$(2r+1)^2 = 3\,(2n+1)^2$$

Now, expanding both sides yields:

$$4r^2 + 4r + 1 = 3\left(4n^2 + 4n + 1\right)$$

$$4r^2 + 4r + 1 = 12n^2 + 12n + 3$$

$$4r^2 + 4r = 12n^2 + 12n + 2$$

$$4r^2 + 4r = 12n^2 + 12n + 2$$

The left side is clearly divisible by 4.

$$r^2 + r = 3n^2 + 3n + \frac{2}{4}$$

The left side is an integer, but the right side is not, which is a contradiction. Therefore our assumption about both p, q not even is false. However, since this can always be done for the rational number p/q it implies that there is no rational p/q such that $(p/q)^2 = 12$.

As a second example, let us take a known true statement and negate the conclusion to see that it will lead to a contradiction. This was inspired by a MathDoctorBob example.

Suppose P is true, and $P \implies Q$ is true. Then by "modus ponens," we have $(P \wedge (P \implies Q)) \implies Q$ is true, so that Q is true.

Now, suppose we wanted to prove that Q is true by contradiction. We will do this proof in a list format (also called "formal") in contrast with the "informal" prose style used above. Our statements will be on the left, and the reason will be on the right.

1.	$\neg Q$	Hypothesis
2.	P	Hypothesis
3.	$P \implies Q$	Hypothesis
4.	$\neg Q \implies \neg P$	Logical equivalence to (3)
5.	$\neg P$	modus ponens (1) and (4)
6.	$\neg P \wedge P$	Contradiction between (2) and (5)

Since we arrived at a contradiction in Step 6, we conclude that our assumption $\neg Q$ must be false and thus Q true. End of proof.

2.3 Direct Proof

The direct proof method is perhaps the conceptually most straightforward method, although it seems, at least "anecdotally" that the other two methods appear as often as direct methods. In any case, we will examine a few illustrations of this approach. Following MathDoctorBob's video, we emphasize that a direct proof is a sequence of true statements (which we can think of as joined together with logical "ands") that logically imply a conclusion. That is to say, that if all the statements are true and the conclusion is true, then we have proved the theorem. Symbollically, this is $H_1 \wedge H_2 \wedge \ldots \wedge H_n \Longrightarrow T$ where T is the conclusion.

We first do an example, following MathDoctorBob, using predicate logic in list format (i.e. formal proof).

We wish to prove that if $P \Longrightarrow Q \wedge R, Q \vee S \Longrightarrow T, P \vee S$ (i.e. each of the statements separated by commas is individually true), then T (is true).

1.	$P \Longrightarrow Q \wedge R$	Hypothesis
2.	$Q \vee S \Longrightarrow T$	Hypothesis
3.	$P \vee S$	Hypothesis
4.	$Q \wedge R \Longrightarrow Q$	Simplification
5.	$P \Longrightarrow Q$	Hypothetical syllogism (1) and (4)
6.	$P \vee S \Longrightarrow Q \vee S$	Constructive dilemma (5)
7.	$Q \vee S$	modus ponens (3) and (6)
8.	T	modus ponens (2) and (7)

We note that statements 4,5, and 6 introduce some new terminology. Furthermore, it may or may not be clear how to proceed from the hypotheses (statements 1,2, and 3) to the conclusion (statement 8). As MathDoctorBob points out, it takes some experience to learn the appropriate rules and patterns, and one must rely on experts to guide a learner through this process. For independent study, reading a variety of proofs, attempting to understand each step, and trying to reproduce them is helpful in this regard. We will do a few exercises to gain a better understanding of the above proof.

Exercise As was mentioned earlier, each statement must be true for the theorem to be proved. Please verify that the statement in step 4 is in fact true by constructing the truth table. Furthermore, what is such a statement called?

Solution We construct the truth table for $Q \wedge R \Longrightarrow Q$ which will require four rows.

$(Q$	$\wedge$	$R)$	$\Longrightarrow$	Q
T	T	T	T	T
T	F	F	T	T
F	F	T	T	F
F	F	F	T	F

We see that this statement is true (i.e. the entries in the column for $\Longrightarrow$) for every possible combination of true and false values of P and Q. This type of statement is called a "tautology."

Exercise Show how true statements (1) and (4) lead to true statement (5).

Solution We first begin by analyzing statement (1) to see under what conditions it is true. Suppose P is true, then Q and R must both be true (recalling the truth table for $\Longrightarrow$). Thus statement (5) $P \Longrightarrow Q$ will also be true. Now suppose P is false. Then there is no restriction on the values of Q and R. Thus, if P is false, no matter what Q is, statement (5) will be true. Therefore we have established statement (5) as being true for all allowed cases. Note, we didn't directly use statement (4) in our analysis.

Exercise Consider statement (6) which is constructed from statement (5) by logical "or" with S on each side. Please show statement (6) is true for all allowed values of P, Q, and S.

Solution We will proceed in a similar approach to the previous exercise. Suppose S is true. Then statement (6) is always true regardless of the values of P and Q. Now suppose S is false. If P is false, then (6) is true no matter what Q is. Suppose P is true. If Q is false, then (6) would be false. However, statement (5), $P \Longrightarrow Q$, prevents this possibility, since if P true then Q true. Therefore, we have verified that (6) is true for all allowed values of the variables P, Q, S. To verify that we didn't overlook a legitimate combination, we may construct a truth table with eight rows:

$(P$	$\vee$	$S)$	$\Longrightarrow$	$(Q$	$\vee$	$S)$
T	T	T	T	T	T	T
T	T	F	T	T	T	F
T	T	T	T	F	T	T
T	T	F	F	F	F	F
F	T	T	T	T	T	T
F	F	F	T	T	T	F
F	T	T	T	F	T	T
F	F	F	T	F	F	F

As we can see, only the entry in row 4, in the column for $\Longrightarrow$ is false. However, this value is not possible because we cannot have P true and Q false, due to statement (5) as was previously mentioned.

At this point, it is useful to summarize what makes a logical argument valid, based on the MathDoctorBob youtube reference (BM3 "Formal Proofs"). The statements in a (formal) proof are constructed from the following list:

- Hypotheses H_i (which are considered true by definition)

- Tautologies
- Substitutions
- Rules of inference using earlier statements P_i (modus ponens is a key example).

We already have some familiarity with statements that are tautologies. Let us examine legitimate substitutions. There are two basic types.

Firstly, consider a statement A that is a tautology which is constructed with another statement S. We are free to replace S by any other statement S^* and obtain a new statement A^* which is also a tautology.

The second type utilizes logical equivalence. Consider a compound statement A with a statement S. We can replace S with a logically equivalent statement S^* and obtain a new statement A^* which is logically equivalent to A.

We consider some exercises to illustrate the substitution rules.

Exercise We previously noted that the statement in step (4) of the proof above was a tautology, namely $Q \wedge R \Longrightarrow Q$. Show that if we substitute $Q \vee R$ for Q, the resulting statement is also a tautology.

Solution Let us construct the truth table of the new statement:

$((Q$	$\vee$	$R)$	$\wedge$	$R)$	$\Longrightarrow$	$(Q$	$\vee$	$R)$
T	T	T	T	T	T	T	T	T
T	T	F	F	F	T	T	T	F
F	T	T	T	T	T	F	T	T
F	F	F	F	F	T	F	F	F

We see the entries in the column for $\Longrightarrow$ are all true, thus the new statement is indeed a tautology.

Let us consider a case where we perform an improper substitution.

Exercise Suppose we consider the same statement as in the previous exercise, $Q \wedge R \Longrightarrow Q$, but this time treat $Q \wedge R$ as the formula we wish to replace with a substitution, R. Show this does not yield a tautology and suggest why the (improper) substitution fails in this case.

Solution If we make the indicated substitution, we obtain $R \Longrightarrow Q$. This is no longer a tautology, since we can have R true and Q false, leading to $R \Longrightarrow Q$ false. As to why this substitution didn't work as expected, we can see that in the original statement, the term on the left of the $\Longrightarrow$ has some relationship with the term on the right, namely through the sharing of Q. The new statement,

$R \Longrightarrow Q$, breaks this connection, thus having a different outcome is not entirely surprising, but perhaps interesting.

Chapter 3

Modern Algebra

Modern algebra is concerned with abstract representations of various mathematical objects, and their manipulations and properties. We will largely be following the material from the book *Algebra* by Michael Artin, as well as some content from Schaum's Outline of Theory and Problems of Abstract Algebra by Frank Ayres, Jr. and Lloyd R. Jaisingh.

The reader may be familiar with some of the topics such as groups from courses in physics, but the emphasis here will be to focus on the pure mathematical aspects, with much attention given to theorems and their proofs. Our treatment will be relatively abbreviated compared to a full course on the subject, but should give the reader enough exposure to feel comfortable with the basics of the subject of abstract algebra.

3.1 Groups

Informally, groups capture the notion of a *symmetry*, which can be thought of as a transformation of an object which leaves it apparently unchanged. For example, consider a square cut from a piece of paper. If we place the square in front of us flat on the table, then rotate it by one-quarter turn, the square will return to its original appearance. We can repeat this operation and observe the same result. We can imagine doing some kind of arithmetic with the square by doing a single rotation, followed by a double rotation, resulting in a sort of addition operation.

To make this more concrete, consider marking each corner of the square by a different letter. Perhaps the top left corner will be a, top right b, bottom right c and bottom left d. Let us define a quarter clockwise turn by the symbol x, a half clockwise turn y, and a three quarter clockwise turn by z. Lastly, let us denote no rotation as e, representing the "identity" symmetry. We can no specify some particular orientations as follows. Consider two x operations, which we will call

x^2 (that is, $x \cdot x$). Now we can see that $x^2 = y$, as an example. We now can consider our "group" to consist of the set of elements $\{x, y, z, e\}$. Each element represents one of the possible operations.

Now, suppose we start with the square with the top left corner a, and we apply the x operation to the square. We will then have the top left corner as d and the top right corner as a. This way we can follow the square through all the symmetries, as each one except e, the identity, will change which letters are in which locations.

To move beyond a rough intuitive concept, let us now more precisely define a group, following Artin.

Definition A *group* is a set G together with a law of composition that has the following properties:

- The law of composition is associative: $(ab)c = a(bc)$ for all $a, b, c \in G$.
- G contains an identity element 1, such that $1a = a$ and $a1 = a$ for all a in G.
- Every element a of G has an inverse, an element b such that $ab = 1$ and $ba = 1$.

The *law of composition* is the manner by which we combine elements of the group to form new objects. Note carefully that the definition refers to a *set*, so we may recall the axioms and properties of set theory from Chapter 1. Also, note that Artin uses 1 for the identity element instead of e. Both are present in different material on groups, but they amount to the same concept.

Exercise It is asserted that the set we introduced earlier to represent the symmetries of the square is a group. How would we prove this?

Solution Let us call the set G, which was defined as $G = \{x, y, z, e\}$. To prove that this set is a group, we would need to verify that it satisfies each criterion in the definition we gave above, namely (1) the composition law is associative, (2) the set contains an identity element, and (3) every element has an inverse.

We have described the law of composition informally, but let us now be more precise. Following Artin, we have the following definition:

Definition A *law of composition* on a set S is any rule for combining pairs a, b of elements of S to get another element, say p, of S. Formally, a law of composition is a function of two variables, or a map

$$S \times S \to S$$

where $S \times S$ is the product set whose elements are pairs a, b of S.

Again, let us note carefully that the law of composition is defined as a <u>*set*</u> (since functions are sets), with a certain structure, namely an ordered pair of elements that are mapped to another element, all from the same set. The composition law is sometimes written as $a \circ b$, where the $\circ$ is a generic symbol to represent the composition. In many cases, we see something more like product or addition, such as ab or $a + b$, but in every case, we must be given the rule that applies to the particular set in question.

Exercise Prove that the identity element for a law of composition on a set S is unique. Use the notation e for identity for convenience, and let $a \in S$ represent elements in S.

Solution We proceed using proof by contradiction. Suppose there are two distinct elements, e and e', in S which satisfy the definition of identity, namely $ea = a$ and $ae = a$ for all $a \in S$. We can therefore write $ee' = e'$ and $ee' = e$. Since the left-hand sides of these two expressions (i.e. ee') are equal, the right-hand sides being unequal is a contradiction. Therefore, we reject the assumption that the elements e and e' are distinct.

Let us now consider a finite set of integers, with addition being the law of composition, and examine it to see if it does or does not form a group. Let the set be $S = \{-3, -2, -1, 0, 1, 2, 3\}$.

Exercise Consider the set S defined above and determine if it meets the definition of a group, under the operation of ordinary integer addition as the law of composition.

Solution Clearly, we have 0 as the identity element, and each element has an inverse (for example $-3 + 3 = 0$ and so forth). However, we cannot properly apply the law of composition because doing so produces elements outside our set. For example, $3 + 2 = 5 \notin S$, therefore we cannot verify associativity holds for all elements of the set since the addition operation is not closed over our set S. (This exercise adapted from Schaums).

Definition The *order* of a group G is the number of elements in the set, and is denoted by $|G|$. We note that there can be groups of finite or infinite order. We will refer to the order of various groups from time to time.

We now continue with a useful law which we will prove in an exercise, adapted from Artin.

Exercise Prove the *Cancellation Law* for groups, which is: given $a, b, c \in G$, where G is a group whose law of composition is written multiplicatively, we have that if $ab = ac$ or $ba = ca$, then $b = c$. Also, if $ab = a$ or $ba = a$, then $b = 1$ (i.e. the identity for the group).

Solution Let us denote the inverse of a as d. Since G is a group, we know the inverse exists. Then we left multiply $ab = ac$ by d, which yields $dab = dac$. Now, $da = 1$ so we end up with $1b = 1c$ which yields $b = c$. We can similarly right-multiply $ba = ca$ by d obtaining $bad = cad$, leading to $b1 = c1$ or $b = c$ again. Lastly, we left multiply $ab = a$ by d yielding $dab = da$ or $1b = 1$, which is $b = 1$, and similarly right multiply $ba = a$ by d, which results in $bad = ad$, or $b1 = 1$ or $b = 1$. We note that the properties of the identity, 1, and the inverse d were stated in the definition of a group earlier, and we freely utilized the associative property of the composition law, again since we have been given a group.

A very useful group is the *symmetric group*, S_n which is the group of the permutations of the indices $1, 2, \ldots, n$. For example, suppose $n = 2$, then we would have the following permutations 12 and 21. Then the group S_2 would be a set containing two elements, namely 1 (the identity) and τ, which represents the transposition operation switching 12 to 21. In other words, $S_2 = \{1, \tau\}$.

Exercise Use the cancellation law for groups to prove that $\tau\tau \neq \tau$, where τ is the transposition permutation in S_2.

Solution Suppose $\tau\tau = \tau$. From the cancellation law for groups, we know that if $ab = a$ for $a, b \in G$, where G is group, then $b = 1$. In our case, we know that $\tau \neq 1$, since the identity is unique, therefore our assumption must be wrong and $\tau\tau \neq \tau$. (Adapted from Artin).

We now introduce the definition of a subgroup (after Artin).

Definition A subset H of a group G is a *subgroup* if it has the following properties:

- *Closure*: If a and b are in H, then ab is in H.
- *Identity*: 1 is in H.
- *Inverses*: If a is in H, then a^{-1} is in H, where a^{-1} means the inverse of a.

We note that a group G is also a subgroup of itself. Furthermore, the subgroup that only contains the element 1 (identity) is called the *trivial subgroup*.

We now do a theorem involving subgroups as an exercise.

Exercise (Adapted from Schaums Abstract Algebra) Prove the following theorem: A non-empty subset G' of a group G is a subgroup of G if and only if for all $a, b \in G'$, $a^{-1}b \in G'$.

Solution Let us first note that in proofs for theorems involving "if and only if", we take the following approach. First we assume the first statement is true then show this implies the second statement, and then we do the reverse. In other words, we have a theorem in the form $A \iff B$, so we first assume A is true, and show that B is true, which means $A \implies B$ is true, and then we assume B is true, then show A is true, giving us $B \implies A$, which then yields $A \iff B$ is true.

Let us assume G' is a subgroup of G. Since $a, b \in G'$, so are all the inverses, by definition of subgroup (inverses property). Then $a^{-1}b \in G'$ by definition of a subgroup (closure property). Thus we have proved $A \implies B$, where A is "G' is a subgroup of G", and B is "for all $a, b \in G', a^{-1}b \in G'$".

Let us now assume for all $a, b \in G', a^{-1}b \in G'$. At this point, all we know about G' is that it is a non-empty subset of the group G, along with $a^{-1}b \in G'$. We have to show this satisfies all the properties required for the definition of a subgroup. Clearly, the identity is in G' since $a^{-1}a \in G'$ and $a^{-1}a = 1$ by definition of identity. Secondly, since we know the identity is in G^{-1}, then so are all the inverses for any $a \in G'$ because $a^{-1}b \in G'$ and if $b = 1$, then we have $a^{-1}1 \in G'$ or $a^{-1} \in G'$ by the property of the identity. Finally, we must show closure. That is we must show that $ab \in G'$ for all $a, b \in G'$. Since we have $a^{-1}b \in G'$, let us substitute a^{-1} for a, yielding $(a^{-1})^{-1}b \in G'$ or $ab \in G'$, thus showing the closure law holds and completing the proof. That is we showed $B \implies A$ is true, where A and B are statements defined in the previous paragraph.

Let us now consider a new group called the "additive group of integers" written as $\mathbb{Z}^+$ in Artin. The composition law for this group is ordinary addition, written as $a + b$, where $a, b \in \mathbf{Z}$. We are interested in subgroups of this group. Per Artin, we say that a subset S of a group G with law of composition written additively is a subgroup if it has the following properties:

- *Closure*: If a and b are in S, then $a + b$ is in S.
- *Identity*: 0 is in S.
- *Inverses*: If a is in S, then $-a$ is in S.

We should recall that a group is a set with a law of composition. In other words, the law of composition is some kind of "attachment" that goes along with the set. We need both component in order to define the group. Therefore, we augment the set of integers $\mathbb{Z}$ with the superscript $+$ sign to indicate the (additive) group, rather than just the set of integers.

The subset of $\mathbb{Z}$ consisting of all multiples of a non-zero integer a is denoted (in Artin) as $\mathbb{Z}a$. We now will prove a related theorem as an exercise.

Exercise Prove the following theorem (Theorem 2.3.3 in Artin): Let S be a subgroup of the additive group $\mathbb{Z}^+$. Either S is the trivial subgroup $\{0\}$, or it has the form $\mathbb{Z}a$, where a is the smallest positive integer in S.

Solution If S is a subgroup of $\mathbb{Z}^+$ then it clearly contains the identity element 0 by the identity property of subgroups. Furthermore, if 0 is the only element, then we are done with the first part of proof, namely we have that S is the trivial subgroup.

Our general strategy for the second part of the proof where subgroups S contain more than the identity element 0 is the following: we will proceed by showing that $\mathbb{Z}a \subseteq S$ and $S \subseteq \mathbb{Z}a$ which means that $\mathbb{Z}a = S$, i.e. the sets are equal, meaning they have the same elements.

Let us first show that $\mathbb{Z}a \subseteq S$, for some positive integer a. Since subgroup S has more than the identity 0, it has at least one non-zero integer n. As S is a subgroup, it must satisfy the properties of a subgroup, therefore we know that S must also contain $-n$ in order to satisfy the inverses property, and furthermore, S must contain a positive number since one or the other of n or $-n$ must be positive. Let us consider the smallest positive number in S and call it a without loss of generality. Clearly, $a + a = 2a$ must also be in S by the closure property of subgroups, and likewise $a + 2a = 3a$ and so on for any positive multiple of a. Also, since a is in S, so is $-a$, and we thus have $-a + (-a) = -2a$ and likewise for any negative multiple of a. Lastly $0a = 0$ is in S. We have satisfied the definition of $\mathbb{Z}a$ and thus we conclude $\mathbb{Z}a \subseteq S$.

Now, we show $S \subseteq \mathbb{Z}a$. Since we know that $\mathbb{Z}a \subseteq S$, this means that any remaining elements of S not in $\mathbb{Z}a$, if they exist, must lie between consecutive multiples of a. Thus, we can write that some arbitrary positive value has the form $p = na + r$, where n is an integer greater than or equal to 1, and integer r is such that $0 \leq r \leq a$. Suppose there was such a p. Then it would possible to write the following sum $p - na$, since $-na$ is also in the group, and this results in $p - na = r$, with r a non-negative integer less than a. However, we know that a is the smallest positive integer in S, therefore this r is a contradiction unless $r = 0$. Therefore, we conclude that $S \subseteq \mathbb{Z}a$ and finally $S = \mathbb{Z}a$ where a is the smallest positive integer in S.

There is an important type of group called a cyclic group. This group is generated by an arbitrary element x of a group G. Following Artin, we use multiplicative notation and define the *cyclic subgroup* H generated by x is the set of all elements that are powers of x. That is $H = \{\ldots, x^{-2}, x^{-1}, 1, x, x^2, \ldots\}$.

The proof of part of the following proposition (theorem) from Artin (Proposition 2.4.2) will be considered in an exercise:

Exercise Let $\langle x \rangle$ be the cyclic subgroup of a group G generated by an element x, and let S denote the set of integers k such that $x^k = 1$. Prove the set S is a subgroup of the additive group $\mathbb{Z}^+$.

Solution In order to prove this, we need to show that set S satisfies the definition of a subgroup of the additive group of integers $\mathbb{Z}^+$. Clearly, the identity $k = 0$ is in S, since $x^0 = 1$ for any element x. We now check for inverses and closure. Suppose k is such that $x^k = 1$, then $x^{-k} = 1$ since $x^{-k} = (x^k)^{-1} = 1^{-1} = 1$. Therefore, S contains inverses. Lastly, to show closure we note that suppose k_1 and k_2 are such that $x^{k_1} = 1$ and $x^{k_2} = 1$ then $x^{k_1+k_2} = 1$ since $x^{k_1+k_2} = x^{k_1}x^{k_2} = 1 \cdot 1 = 1$ by properties of exponents, so

$k_1 + k_2$ is also in S.

We now discuss *homomorphisms*, following Artin.

Definition A *homomorphism* $\phi : G \to G'$, where G, G' are groups, is a map from G to G' such that for all $a, b \in G$, $\phi(ab) = \phi(a)\phi(b)$.

A good example of such a homomorphism is the "exponential map": $\exp : \mathbb{R}^+ \to \mathbb{R}^\times$, defined by $x \to e^x$, where $\mathbb{R}^+$ is the set of real numbers, with addition as the law of composition, $\mathbb{R}^\times$ is the set of nonzero real numbers, with multiplication as the law of composition, and e is the usual base of the natural logarithm. In particular, let $a, b \in \mathbb{R}^+$, then we have $e^{a+b} = e^a e^b$ following the usual law of exponents.

Let us consider the following proof as an exercise (adapted from Artin, Proposition 2.5.3).

Exercise Let $\phi : G \to G'$ be a group homomorphism. Show that ϕ maps the identity in group G to the identity in group G'.

Solution Consider applying the homomorphism to $1a$ where 1 is the identity in G and $a \in G$. We thus obtain $\phi(1a) = \phi(1)\phi(a)$. Since $\phi(1a) = \phi(a)$, because $1a = a$, we further have that $\phi(1a) = \phi(1)\phi(a) = \phi(a)$. Likewise, we have that $\phi(a1) = \phi(a)\phi(1) = \phi(a)$, since $a1 = a$, and $\phi(a1) = \phi(a)$. Thus, $\phi(1)$ has the properties of the identity, and we conclude it is the identity of G'.

We now introduce some important additional terminology (following Artin) related to groups.

- The *image* of a homomorphism.
- The *kernel* of a homomorphism.
- The *coset* of a group.
- A *normal subgroup* of a group.
- The *center* of a group.

Let us consider each of the above concepts.

The *image* of homomorphism ϕ is $\mathrm{im}\phi = \{x \in G' | x = \phi(a) \text{ for some } a \in G\}$, where we have $\phi : G \to G'$, with G, G' groups.

The concept of a *kernel* appears frequently, and should be kept in mind. It is the set of elements of a group G that are mapped to the identity of G' by the homomorphism. That is, $\ker\phi = \{a \in G | \phi(a) = 1\}$, where 1 is the identity of G'.

We do a proof exercise to illustrate the kernel as a subgroup.

Exercise Prove the kernel of the homomorphism $\phi : G \to G'$ is a subgroup of G. (Adapted from Artin).

Solution We must show that the kernel $\ker\phi$ meets all the properties of a subgroup, namely, closure, identity, and inverses. First, we verify closure. Suppose a and b are in the kernel, i.e., a, b are in the subset of G that ϕ maps to the identity of G'. Since ϕ is a homomorphism, we have that $\phi(a)\phi(b) = \phi(ab) = 1$, since $\phi(a)\phi(b) = 1 \cdot 1 = 1$. Thus ab must also be in the kernel and so we have verified closure.

To verify that the identity is in the kernel, we must have that $\phi(1_G) = 1$ where we have emphasized that 1_G is the identity element of group G by using the subscript G. Suppose a is in the kernel. Then we have $a = 1_G a$ by property of the identity. Thus we have $1 = \phi(a) = \phi(1_G a) = \phi(1_G)\phi(a)$, with the last equality true since ϕ is a homomorphism. Since we have $1 = \phi(1_G)\phi(a)$, we obtain $1 = \phi(1_G) \cdot 1$, and we know by the cancellation law of groups, $\phi(1_G) = 1$, which means 1_G is also in the kernel.

Lastly, we must check for inverses. Suppose that a is in the kernel. We then have $aa^{-1} = 1_G$, where a^{-1} is the inverse of a. Since ϕ is a homomorphism and we know that the identity 1_G is in the kernel, we have that $1 = \phi(1_G) = \phi(aa^{-1}) = \phi(a)\phi(a^{-1})$. We then have $1 = \phi(a)\phi(a^{-1})$ and since a is in the kernel, we have $1 = 1 \cdot \phi(a^{-1})$. Again, by the cancellation law of groups, we conclude the $\phi(a^{-1}) = 1$, and therefore a^{-1} must also be in the kernel.

We now discuss the concept of *coset* of a group. We will speak in some detail about "left cosets," but note that "right cosets" also exist with a similar discussion, but are not necessarily equal to left coset. Following Artin, we make the following definition for a left coset:

Definition If H is a subgroup of G and if a is an element of G, the following subset is called a *left coset*

$$aH = \{ah | h \in H\}$$

where ah is a product (i.e. application of law of composition) of the element a with h. One can think of the left cosets as dividing up, or partitioning, the group G into separate "equivalence classes." We briefly introduced the concept of equivalence (relations) and partitions in Chapter 1. We will now consider equivalence relations in the context of groups.

Definition (from Artin) The cosets of H in G, where H is a subgroup of group G, are called *equivalence classes* for the congruence relation

$$a \equiv b, \text{ if } b = ah \text{ for some } h \in H.$$

We use the notation $a \equiv b$ to indicate the congruence relation between a and b. We recall (from Chapter 1) that a partition of a set is an exhaustive, mutually exclusive decomposition of a set into subsets, whose union is the set itself.

Exercise Prove that the congruence relation defined above is indeed an equivalence relation, in that it satisfies the reflexive, symmetric, and transitive properties.

Solution Let us consider reflexive first. This means that $a \equiv a$. This is clearly satisfied by the definition as follows. Let $b = a \cdot 1$, where 1 is the identity in H, which exists because H is a subgroup. Then we have $a \equiv b$ yielding $a \equiv a \cdot 1$, which leads to $a \equiv a$.

Now, the symmetric property. This means that $a \equiv b \implies b \equiv a$ and $b \equiv a \implies a \equiv b$. Suppose $a \equiv b$. This means that $b = ah$ for some h in H. Right multiplying both sides by h^{-1}, which is the inverse of h and is also in H since H is a subgroup, we get $bh^{-1} = a$. This implies $b \equiv a$.

Last, we wish to prove transitivity. This means that if $a \equiv b$ and $b \equiv c$, then $a \equiv c$, where $a, b, c \in G$. Suppose $a \equiv b$. This means $b = ah$ for some $h \in H$. Now, suppose $b \equiv c$. This means that $c = bh'$, where $h' \in H$. Note, we have a prime on the second h to distinguish it from the first. We now, have that $c = (ah)h'$ by combining the two results. By associativity of the law of composition we have $c = a(hh')$, which implies $a \equiv c$ since hh' is also in H because H is a subgroup and thus is closed.

We note that all left cosets of a subgroup H of a group G have the same order (i.e. same number of elements) and this results in something called the "Counting Formula" which is:

$$|G| = |H|[G : H]$$

where $|G|$ is the order of G, $|H|$ is the order of H and $[G : H]$ is the number of cosets of H in G. Recall that the cosets partition G.

Exercise Prove the order of an element of a finite group divides the order of the group (From Artin). Note: The order of an *element* x of a group is the smallest positive integer n such that $x^n = 1$. This is the same thing as saying that the cyclic subgroup $\langle x \rangle$ generated by x has order n.

Solution Since the order of the element is the same as the order of the corresponding cyclic subgroup, we can use the counting formula. Letting H be the subgroup of a given (finite) group G, and since we know $|G| = |H|[G : H]$, clearly H divides G. We note that when the term "divides" in used in these contexts, integer division is meant. In other words, each term $|G|$, $|H|$, and $[G : H]$ is a positive integer, so of course $|G|/|H|$ is an integer.

Following Artin, we define a *normal subgroup* N of a group G in the following manner: if for every $a \in N$ and $g \in G$, the "conjugate" $gag^{-1} \in N$, where gag^{-1} is the product of the element g, a, and g^{-1}, the inverse of g, obtained by application of the usual law of composition for the group G.

We will do a proof (from Artin) illustrating the normal subgroup.

Exercise Show the kernel of a homomorphism is a normal subgroup.

Solution We first recall that the kernel of a homomorphism is the set of elements of one group G which map to the identity of a second group G' via the homomorphism ϕ. We previously showed that the kernel of a homomorphism is a subgroup, so now we must show it is a *normal* subgroup. Let $g, g^{-1} \in G$ and $a \in K$, where K is the kernel. Now, if $\phi(gag^{-1}) = 1_{G'}$, then K is a normal subgroup. Since ϕ is a homomorphism, $\phi(gag^{-1}) = \phi(g)\phi(a)\phi(g^{-1}) = \phi(g)1_{G'}\phi(g^{-1}) = \phi(g)\phi(g^{-1}) = \phi(gg^{-1}) = \phi(1) = 1_{G'}$. The last equality is due to the fact the kernel is a subgroup and thus contains the identity 1 (of G) and therefore $\phi(1) = 1_{G'}$.

We now describe the *center* of a group G, following Artin. The center, denoted by Z in Artin, is the set of elements of G that commute with every element of G:

$$Z = \{z \in G | zx = xz \forall x \in G\}$$

We now define an important concept called an *isomorphism*, following Artin.

Definition An *isomorphism* $\phi : G \to G'$, from a group G to a group G' is a bijective group homomorphism. Bijective means a unique element in G is mapped to a unique element in G' (injective) and furthermore, every element in G' is mapped from a corresponding element in G (surjective). Another way to say this is that every element in G' is the *image* of some element in G.

However, before continuing our investigation of isomorphisms, let us discuss in some detail the notion of maps and inverse maps. This is somewhat of a review of the topic first introduced in Chapter 1.

To begin with, let us emphasize what is meant by a "map." Following *Schaum's Outline of Theory and Problems of Abstract Algebra* by Ayres and Jaisingh, we can consider a map to be a subset of the product set $G \times G'$. Considering individual elements of this product set as ordered pairs, we may write $G \times G' = \{(g_1, g_1'), (g_2, g_1'), \ldots, (g_m, g_n')\}$, where $g_i \in G$ and $g_j' \in G'$, m is the order of G, and n is the order of G' (we are considering finite groups in this illustration).

Now, to qualify as a "map," the subset must be such that for each unique element of G that appears in the first position of the ordered pair, only one element of G' can be associated with that element of G. For example, $\{(g_1, g_2'), (g_2, g_2')\}$ is allowed, but $\{(g_1, g_2'), (g_1, g_3')\}$ is not allowed. Furthermore, we require that each and every element of G appears in the subset (of course in the first position of the ordered pair).

If we consider the "identity map," $\mathcal{I}$, which maps an element to itself, we can write $\phi\phi^{-1} = \phi^{-1}\phi = \mathcal{I}$, where ϕ is the map $\phi : G \to G'$ and $\phi^{-1} : G' \to G$. If such a ϕ^{-1} exists for a given ϕ, we call ϕ^{-1} the "inverse mapping" of ϕ. Thus, applying the map ϕ followed by the map ϕ^{-1} to a given element results in the same element. In terms of subsets of a product set, the identity map would be a subset of $G \times G$, and we might indicate the subset as $\mathcal{I} : G \to G$. An example might be $\mathcal{I} = \{(g_1, g_1), (g_2, g_2)\}$. When we write $\phi\phi^{-1}$, we really mean the composition of ϕ with ϕ^{-1}. Since each map ϕ and ϕ^{-1} are subsets of product sets, we need to have some way to construct the composition itself, which is also a mapping. Thus, we identify a rule, in this case $s' = \phi^{-1}(\phi(s))$ or $s' = \phi(\phi^{-1}(s))$, which says that the first element in the ordered pair is s and the second element is s' obtained by applying ϕ^{-1} to the result of $\phi(s)$ in the first case, or applying ϕ to the result of $\phi^{-1}(s)$ in the second. This rule is unambiguous and yields the desired subset of $G \times G$.

There is no guarantee that a given map ϕ will have a corresponding inverse in the manner described above. For example, consider $\phi = \{(g_1, g_2'), (g_2, g_2')\}$. Since both g_1 and g_2 are mapped by ϕ to g_2', there is no inverse map which will result in the identity mapping as described above. However, if we had something like $\phi = \{(g_1, g_2'), (g_2, g_1')\}$, then an inverse mapping exists. In fact, we can easily obtain it by swapping the entries in each ordered pair: $\phi^{-1} = \{(g_2', g_1), (g_1', g_2)\}$. There is obviously something to be said for the utility of unique mappings from the domain to co-domain. These are also called one-to-one mappings, which we introduced in Chapter 1.

In contrast, we can have one-to-one mappings between two sets which are not inverses. For example consider $\phi = \{(g_1, g_2'), (g_2, g_1')\}$ and $\gamma = \{(g_1', g_1), (g_2', g_2)\}$ where $g_i \in G$ and $g_j' \in G'$.

It is a theorem given in Schaum's (Theorem I) that if α is a one-to-one mapping of a set S *onto* a set T, then α has a unique inverse, and conversely. We emphasize that one-to-one means that images of distinct elements of the domain, S in this case, are distinct elements of the co-domain, T. We previously called this "injective." The term "onto" means that every element of T is the image of some element in S. This is also called "surjective." We now present the proof of Schaum's Theorem I.

We must first show that an inverse map exists, then we will show it is unique.

Since we know that α is one-to-one, this means that $\alpha(s) = t$ where t is distinct for every distinct s. Furthermore, since α is onto, this means that every t is the image of some s. Thus, we can define an inverse map, α^{-1}, with the property that if $\alpha(s) = t$, then $s = \alpha^{-1}(t)$ for all $t \in T$. We briefly verify the α^{-1} is a map. First of all, it is defined for every $t \in T$, and secondly, for every distinct t we have a corresponding distinct s. Thus, each distinct t only appears once in each distinct ordered pair as the first element.

Moving on to uniqueness, let us call the inverse map to α, β, and let us now suppose a second distinct inverse to α, call it γ, exists. Then we can write $\beta\alpha = \gamma\alpha = \mathcal{I}$. By right multiplying each term by β we see that $\gamma\alpha\beta = \mathcal{I}\beta$ which shows that $\gamma = \beta$ contradicting our assumption that γ is distinct from β. Therefore, the inverse is unique. We can also write $\alpha\beta = \alpha\gamma = \mathcal{I}$ and left multiply by β which results in the same conclusion, namely $\beta\alpha\beta = \beta\alpha\gamma = \beta\mathcal{I}$ or $\beta = \gamma$.

Now suppose the converse. That is, suppose α has a unique inverse, then α is a one-to-one map S onto T. Let β be the unique inverse to α. We proceed using proof by contradiction. Suppose that α is not a one-to-one map from S to T. This means that images of distinct elements of the domain S are not distinct elements of the co-domain T. Let $a, b \in S$ be distinct elements such that $\alpha(a) = \alpha(b)$. Since we are given that β is the unique inverse to α, we obtain $\beta(\alpha(a)) = \beta(\alpha(b))$ which leads to $a = b$, clearly a contradiction.

Now suppose that the mapping α is not onto. That means that there is some element in T that is not an image of some element in S. However, since we are assuming that α has a unique inverse, each element of T must map to an element of S. Call the inverse β, and let $t \in T$. Then we have $s = \beta(t)$, where $s \in S$. Applying the forward map to both sides yields $\alpha(s) = \alpha(\beta(t)) = t$, thereby showing that t is the image of $\alpha(a)$ contradicting our supposition, therefore showing that α is indeed onto. This concludes the proof of Schaum's Theorem I.

With the above discussion of maps and inverse maps now completed, we can further investigate isomorphisms in the following exercise.

Exercise Prove that if $\phi : G \to G'$ is an isomorphism, the inverse map $\phi^{-1} : G' \to G$ is also an isomorphism (Adapted from Artin).

Solution Since ϕ is an isomorphism, by definition it is a bijective (i.e. both injective and surjective) group homomorphism. We recall that a group homomorphism is a map from one group to another. To complete the proof, we must show that

- The inverse map, ϕ^{-1}, is a group homomorphism.
- The inverse map, ϕ^{-1}, is injective.
- The inverse map, ϕ^{-1}, is surjective.

We now continue with proving the three necessary properties of the inverse map ϕ^{-1} to show it is indeed an isomorphism:

Proof that the inverse map, ϕ^{-1}, is a group homomorphism:

Since we are given that ϕ is an isomorphism, we know that it is a one-to-one, onto map. The theorem we just proved above (Schaum's Theorem I) tells us that there exists a unique inverse map.

We should reiterate what it means to be a homomorphism. In other words, what are the defining properties in the context of mappings. Recalling the definition from earlier: A *homomorphism* $\phi : G \to G'$, where G, G' are groups, is a map from G to G' such that for all $a, b \in G$, $\phi(ab) = \phi(a)\phi(b)$.

Therefore, we know that ϕ^{-1} exists and is unique. Applying it to the defining relation above yields $\phi^{-1}(\phi(ab)) = \phi^{-1}(\phi(a)\phi(b)) = ab = \phi^{-1}(\phi(a))\phi^{-1}(\phi(b))$, since $\phi^{-1}(\phi) = \mathcal{I}$ by definition of inverse map. This shows that ϕ^{-1} is a homomorphism, because $\phi^{-1}(\phi(a)\phi(b)) = ab = \phi^{-1}(\phi(a))\phi^{-1}(\phi(b))$ for all $\phi(a), \phi(b) \in G'$, which is the definition of homomorphism.

Proof that the inverse map, ϕ^{-1}, is injective:

We first prove that a homomorphism is injective if and only if its kernel is the trivial subgroup (Corollary 2.5.9 in Artin). That is, $\phi : G \to G'$ is injective $\iff$ the kernel $K = \{1\}$.

Suppose ϕ is injective. This means that a unique element of G maps to a unique element of G'. If the kernel exists, then it must contain at least the identity since it is a subgroup. We know that the identity is unique, therefore since ϕ is injective, it must map the identity of G to the identity of G', by definition of kernel. Furthermore, the identity of G is the only element in the kernel, and therefore is the trivial subgroup $\{1\}$ of G by definition.

Now suppose that the kernel K of homomorphism ϕ is the trivial subgroup of G. We proceed using proof by contradiction, in particular, assuming that ϕ is not injective. Let $a, b \in G$ be two distinct elements. Now, suppose $\phi(a) = \phi(b)$. This means that $[\phi(a)]^{-1}\phi(a) = 1_{G'} = [\phi(a)]^{-1}\phi(b)$, by definition of inverse. We note that $\phi(1) = \phi(aa^{-1}) = \phi(a)\phi(a^{-1}) = 1_{G'}$ since ϕ is a homomorphism and we are supposing that the kernel of ϕ is the trivial subgroup $\{1\}$. This means that $\phi(a^{-1}) = [\phi(a)]^{-1}$, obtained by left multiplication of $\phi(a)\phi(a^{-1}) = 1_{G'}$ by $[\phi(a)]^{-1}$.

Thus, we have $1_{G'} = [\phi(a)]^{-1}\phi(b) = \phi(a^{-1})\phi(b) = \phi(a^{-1}b)$. Therefore, $a^{-1}b$ is in the kernel and in this case since the kernel is $\{1\}$ we have $a^{-1}b = 1$, so that leads to $aa^{-1}b = a1$ or $b = a$, which contradicts our assumption that a was distinct from b. Therefore, $\phi(a) \neq \phi(b)$ and we have proved that a homomorphism is injective if and only if its kernel is the trivial subgroup.

We now need to show that the kernel of ϕ^{-1} is the trivial subgroup $\{1_{G'}\}$.

Suppose $a \in K'$ where K' is the kernel of ϕ^{-1}. Then, $1_G = \phi^{-1}(a)$. Now, $\phi(1_G) = \phi(\phi^{-1}(a)) = 1_{G'}$. Since $\phi(\phi^{-1}(a)) = a$, we have $a = 1_{G'}$. Thus, since the kernel of ϕ^{-1} is the trivial subgroup, ϕ^{-1} is injective, by the theorem proved above (Corollary 2.5.9 in Artin).

Proof that the inverse map, ϕ^{-1}, is surjective:

We have to show that every element of S is the image of some element of T. Since we are assuming that we have an inverse map, call it ϕ^{-1}, we can apply $\phi(a)$ to some arbitrary element $a \in S$, yielding $t = \phi(a)$. Then applying the inverse map to both sides, we obtain $a = \phi^{-1}(t)$, showing that a general element a is the image of $\phi^{-1}(t)$.

This completes the proof that if $\phi : G \to G'$ is an isomorphism, the inverse map $\phi^{-1} : G' \to G$ is also an isomorphism.

3.2 Rings

We now discuss a new abstract object called a *Ring*. The set of integers $\mathbb{Z}$ is an example of a ring. We begin by describing the defining properties of rings. We note that groups are probably familiar to students of physics certainly, but rings not as likely. However, we will see that the development is similar in general structure, so being familiar with groups definitely provides a good foundation for the continued investigation.

Definition (From Artin). A *ring* R is a set with two laws of composition $+$ and $\times$, called addition and multiplication, that satisfy these axioms:

- With the law of composition $+$, R is an abelian group that we denote by R^+; the group identity is denoted by 0. We note that "abelian" means that $a + b = b + a$ if $a, b \in R^+$, that is, the order does not matter.
- Multiplication is commutative and associative, and has an identity denoted by 1.
- *distributive law*: For all a, b, and c in R, $(a + b)c = ac + bc$.

We can immediately see that the set of integers, with the ordinary definitions of $+$ and $\times$ satisfies these axioms.

Artin defines a *subring* of a ring as a subset that is closed under the operations of addition, subtraction, and multiplication and that contains the element 1.

Further discussed in Artin, there is a special ring with just one element 0, called the *zero ring*. We explore this in the following exercise (Adapted from Artin).

Exercise Prove that a ring R in which the elements 1 and 0 are equal is the zero ring.

Solution: Proceeding using proof by contradiction. Suppose our ring has (equal) elements $1, 0$ and at least one other general element, a, that is different from 1 or 0. We must show that trying to satisfy the axioms is not possible unless a is 0 or 1. In order to proceed, we need to clarify what is meant by $1 = 0$ in this context. This means that there is one and only one element of the ring, R, that serves as both the identity for the abelian group R^+, with law of composition $+$, and the identity for the ring R with the law of composition $\times$.

Consider the distributive law axiom for rings, with the above definitions: $(a+1)1 = a1 + 1 \cdot 1 = a + 1$. On the other hand, we have $(a+0)0 = a0 + 0 \cdot 0 = 0 + 0 = 0$. Since are given that $1 = 0$, the results of these two expression must be equal, namely $a + 1 = 0$. Again, since $1 = 0$, we must have $a = 0$. Therefore, since the contradicts our assumption that a is different from zero, $a = 0$.

Following Artin, let us introduce the idea of a *polynomial ring*, which is a set of polynomials with coefficients $a_i \in R$:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

In typical engineering and science applications, one doesn't restrict coefficients to elements of rings, so this is an interesting departure. We can informally observe that such a structure satisfies the axioms of a ring as the ring composition laws are the familiar arithmetic operations of addition and multiplication.

Exercise (Adapted from Artin, Proposition 11.2.9) Let R be a ring, let f be a monic polynomial and let g be any polynomial, both with coefficients in R. Prove that there are uniquely determined polynomials q and r in $R[x]$ such that

$$g(x) = f(x)q(x) + r(x)$$

and such that the remainder r, if it is not zero, has degree less than the degree of f. Artin hints that the proof follows the algorithm for division of polynomials that "one learns in school."

Solution: Firstly, a monic polynomial is one whose leading coefficent, i.e. the term with the highest power of x, is 1. Let us consider the arbitrary polynomial $g(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0$. We know the form of f to be $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$.

We can then obtain $q(x)$ and $r(x)$ by polynomial division, since we know that f is monic and thus has at least one nonzero coefficient, namely the leading coefficient. So we have:

$$\frac{g(x)}{f(x)} = \frac{b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0}{x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}$$

If we divide $g(x)$ by the leading term of $f(x)$ we obtain,

$$b_m x^{m-n} + b_{m-1}x^{m-1-n} + \cdots + b_1 x^{1-n} + b_0 x^{-n}$$

However, before proceeding further, it would be useful to consider the overall structure of the operation we are trying to perform. Let us propose some candidate polynomials $q(x), r(x)$, and see what constraints there are on the formulation.

Suppose $q(x) = c_s x^s + c_{s-1}x^{s-1} + \cdots + c_1 x + c_0$ and $r(x) = d_t x^t + d_{t-1}x^{t-1} + \cdots + d_1 x + d_0$.

Then, we have:

$$\begin{gathered} b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0 = \\ (x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0)(c_s x^s + c_{s-1}x^{s-1} + \cdots + c_1 x + c_0) + d_t x^t + d_{t-1}x^{t-1} + \cdots + d_1 x + d_0 \end{gathered} \tag{3.1}$$

Firstly, let us concentrate on expanding the multiplicative term $f(x)q(x)$:

$$\begin{gathered} f(x)q(x) = \\ c_s x^{n+s} + c_{s-1}x^{n+s-1} + c_{s-2}x^{n+s-2} + \cdots + c_1 x^{n+1} + c_0 x^n + \\ a_{n-1}c_s x^{n+s-1} + a_{n-1}c_{s-1}x^{n+s-2} + \cdots + a_{n-1}c_1 x^n + a_{n-1}c_0 x^{n-1} + \\ a_{n-2}c_s x^{n+s-2} + a_{n-2}c_{s-1}x^{n+s-3} + \cdots + a_{n-2}c_1 x^{n-1} + a_{n-2}c_0 x^{n-2} + \cdots \end{gathered} \tag{3.2}$$

Now collecting terms we obtain:

$$\begin{gathered} c_s x^{n+s} + (c_{s-1} + a_{n-1}c_s)x^{n+s-1} + \\ (c_{s-2} + a_{n-1}c_{s-1} + a_{n-2}c_s)x^{n+s-2} + \\ (c_{s-3} + a_{n-1}c_{s-2} + a_{n-2}c_{s-1} + a_{n-3}c_s)x^{n+s-3} + \cdots + a_0 c_0 \end{gathered} \tag{3.3}$$

We point out there are $n+s+1$ terms which yields $n+s+1$ equations in $s+1$ unknowns, c_s. To be clear, we consider the a_i to be given. Now, let us consider several cases regarding the order of the left-hand side.

Case 1, $m = n$:

If we choose $s = 0$, this provides terms on the right-hand side of the following form:

$$c_0x^m + (a_{m-1}c_0)x^{m-1} + (a_{m-2}c_0)x^{m-2} + (a_{m-3}c_0)x^{m-3} + \cdots + a_0c_0 \quad (3.4)$$

We note that c_s for $s < 0$ is zero. If we set $c_0 = 1$, we can satisfy the equation term-by-term, by using $r(x) = d_tx^t + d_{t-1}x^{t-1} + \cdots + d_1x + d_0$, fixing $t = m$. Then, we have terms of the general form $b_i = a_i + d_i$, where clearly d_m is uniquely defined for each m. However, in order to satisfy the proof statement, we must have $t < n$, so that requires $c_0 = b_m$ and consequently a modification of the equation given earlier to now become $b_i = c_0a_i + d_i = b_ma_i + d_i$ for the $i'th$ order term.

Case 2, $m > n$:

In this case, we choose $s = m - n$ and again fix $t = m$. This results in terms of the following general form:

$$c_{m-n}x^m + (c_{m-n-1} + a_{n-1}c_{m-n})x^{m-1} + (c_{m-n-2} + a_{n-1}c_{m-n-1} + a_{n-2}c_{m-n})x^{m-2} + (c_{m-n-3} + a_{n-1}c_{m-n-2} + a_{n-2}c_{m-n-1} + a_{n-3}c_{m-n})x^{m-3} + \cdots + a_0c_{m-n} \quad (3.5)$$

We see that there are $m+1$ equations, however we have to be cautious in accounting for the number of unknowns. In particular, considering the c_i, there are $m-n+1$ unknowns. Taking the difference, we have $m+1-m+n-1 = n$ variables that must be provided by the $r(x)$ term. If we again set $c_{m-n} = 1$, we will have a typical term, say for x^{m-3}, like:

$$b_{m-3} = c_{m-n-3} + a_{n-1}c_{m-n-2} + a_{n-2}c_{m-n-1} + a_{n-3}$$

We then add in the $r(x)$ term:

$$b_{m-3} = c_{m-n-3} + a_{n-1}c_{m-n-2} + a_{n-2}c_{m-n-1} + a_{n-3} + d_{m-3}$$

One can see that for any choice of coefficients, c_i, there will be an independent degree of freedom, d_j such that the above equation is satisfied. However, again to satisfy the proof statement, $t < n$. As we can see in the above equation d_{m-3} is superfluous, so we are free to set it to zero. We don't require any contribution until the c coefficients are zero. This occurs when the index of c_i is negative, i.e. $i < 0$. Since there are n variables being provided by the $r(x)$ term, this means that the order of $r(x)$ must be $n-1$ which is less than n, since the first index of $r(x)$ is zero. In other words, d_0 is the lowest order term in $r(x)$.

Case 3, $m < n$:

Let us consider this case following a similar approach to the above. We are restricting ourselves to non-negative integer powers, so we need to properly account for this to keep the equation valid.

Considering our general product term again, we have

$$\begin{aligned} c_s x^{n+s} + (c_{s-1} + a_{n-1}c_s)x^{n+s-1} + \\ (c_{s-2} + a_{n-1}c_{s-1} + a_{n-2}c_s)x^{n+s-2} + \\ (c_{s-3} + a_{n-1}c_{s-2} + a_{n-2}c_{s-1} + a_{n-3}c_s)x^{n+s-3} + \cdots + a_0 c_0 \end{aligned} \tag{3.6}$$

We can set $s = 0$, the minimum non-negative value, and then arrange things such that all the coefficients of powers of x where $n > m$, are set to zero.

We first set $s = 0$, yielding:

$$\begin{aligned} c_0 x^n + a_{n-1}c_0 x^{n-1} + \\ a_{n-2}c_0 x^{n-2} + \\ a_{n-3}c_0 x^{n-3} + \cdots + a_0 c_0 \end{aligned} \tag{3.7}$$

Recalling that negative indices of c_s are set to zero, we have removed them. Once again, setting c_0 to 1, we obtain:

$$\begin{aligned} x^n + a_{n-1}x^{n-1} + \\ a_{n-2}x^{n-2} + \\ a_{n-3}x^{n-3} + \cdots + a_0 \end{aligned} \tag{3.8}$$

We now set $t = n$ and add in the terms corresponding to $r(x)$:

$$(1 + d_n)x^n + (a_{n-1} + d_{n-1})x^{n-1} + \\ (a_{n-2} + d_{n-2})x^{n-2} + \\ (a_{n-3} + d_{n-3})x^{n-3} + \cdots + (a_0 + d_0) \quad (3.9)$$

Clearly, the d_i can be chosen the satisfy the below equation, and in fact these choices would be unique:

$$g(x) = f(x)q(x) + r(x)$$

In particular, if b_j is some coefficient of $g(x)$, then $d_j = b_j - a_j$. However, as above, we need to adjust things slightly to satisfy the conditions of the proof. Firstly, c_0 can be set to zero to remove the x^n term, then the remaining terms can be set by appropriate values for d_j. We note that the order of $r(x)$ is less than n under these conditions. This concludes the proof.

For completeness, it is useful to consider what the polynomial division algorithm looks like in practice. That is, suppose we are given $g(x)$ and $f(x)$, how to determine $q(x)$ and $r(x)$, now that we see they exist and can be determined uniquely.

The $m = n$ case is reasonably straightforward. We consider the $m > n$ and $m < n$ cases instead.

Suppose $m > n$. From our analysis, we see that the coefficient c_{m-n} of $q(x)$ is simply the coefficient of the x_m term in $g(x)$. We may proceed down the powers of x^i in this manner, since at each step we introduce a single new variable. When the index $m - n - i$ is negative, we consider the c coefficients to be zero, and begin utilizing the coefficients from the $r(x)$ term to match the coefficients from the $g(x)$ terms. The algorithm terminates at the constant term, when $m - i = 0$.

Likewise, we now suppose $m < n$. Here we set $q(x) = 1$ and use $r(x)$ to satisfy the equation $g(x) = f(x)g(x) + r(x)$. This is done term by term, where as mentioned above, $d_j = b_j - a_j$ in general.

Let us continue our exploration of rings and discuss *ring homomorphisms* and *ideals*.

We discussed group homomorphisms previously, and now we present the definition of a ring homomorphism.

Definition (From Artin) A *ring homomorphism* $\phi : R \longrightarrow R'$ where R, R' are rings, is a map such that:

$$\phi(a + b) = \phi(a) + \phi(b), \quad \phi(ab) = \phi(a)\phi(b), \quad \phi(1_R) = 1_{R'}$$

where we have emphasized the multiplicative identity 1_R is that for the ring R, and the multiplicative identity $1_{R'}$ is the one in ring R'.

Exercise: (Adapted from Artin). The map $\phi : \mathbb{Z} \longrightarrow \mathbb{F}_p$ sends an integer to its congruence class modulo p. Prove this is a ring homomorphism.

Solution: We first describe what a congruence class $\mathbb{F}_p$ is, and then verify each of the properties for a ring homomorphism. We take for granted that both $\mathbb{Z}$ and $\mathbb{F}_p$ are indeed rings or can be treated as rings for this problem.

We take a slight detour to discuss modular arithmetic in order to introduce congruence classes. This is a topic in number theory, but has a place in abstract algebra. Following Artin, we make the following definition:

Definition Two integers a and b are said to be *congruent modulo n*, written as $a \equiv b \mod n$, if n divides $b - a$, where n is a fixed positive integer. We emphasize that the meaning of "divides" here is that there is no remainder. For example, 3 divides 6, but 3 does not divide 7.

With the above definition in mind, we can now define the congruence class of an integer:

Definition The set given by

$$\overline{a} = \{\ldots, a - n, a, a + n, a + 2n, \ldots\}$$

is called the *congruence class of an integer a modulo n.*

Following Artin, if a and b are integers, $\overline{a} = \overline{b}$ means that $a \equiv b \mod n$.
We make the further following definitions for the sum and product of congruence classes of integers.

Definition The sum of the congruence classes of two integers, $\overline{a}$ and $\overline{b}$ is given by $\overline{a} + \overline{b} = \overline{a + b}$, and the product is given by $\overline{a}\overline{b} = \overline{ab}$. It should be reminded that the over line indicates the set representing the congruence class.

We state without proof that the associative, commutative, and distributive laws hold for addition and multiplication of congruence classes, per Artin. Further, we denote the set of congruence classes modulo n, as $\mathbb{Z}/\mathbb{Z}n$. For example, the set of congruence classes for $n = 3$, would be given by $\mathbb{Z}/\mathbb{Z}3 = \{\overline{0}, \overline{1}, \overline{2}\}$, where each of the elements are themselves sets. It's worth verifying this example quotient group is a subgroup of $\mathbb{Z}$, by checking the subgroup properties are satisfied. We briefly do this now.

We consider the group law of composition to be addition. Thus, for the integers, this group is denoted as $\mathbb{Z}^+$, but we retain the original notation, but keeping the additional law in mind. Certainly $\mathbb{Z}/\mathbb{Z}3$ is a subset of $\mathbb{Z}$.

We first verify the identity is in the subgroup. Clearly, the element $\overline{0}$ satisfies the requirements for the identity: $\overline{0}+\overline{a}=\overline{0+a}=\overline{a}=\overline{a+0}=\overline{a}+\overline{0}$, where $\overline{a}$ is any element of $\mathbb{Z}/\mathbb{Z}3$, and we used the definition of sum of congruence classes given above, and the the fact the $\mathbb{Z}$ is an abelian group.

We next verify closure. We only need to check the elements $\overline{1}$ and $\overline{2}$, since we already established $\overline{0}$ is the identity, and any sum involving it will definitely be closed. $\overline{1}+\overline{1}=\overline{2}$ which is closed. $\overline{1}+\overline{2}=\overline{3}=\overline{0}$ since we are in modulo 3, thus this is closed. $\overline{2}+\overline{1}$ yields the same conclusion. Finally $\overline{2}+\overline{2}=\overline{2+2}=\overline{4}=\overline{1}$ since we in modulo 3 as stated earlier. Thus closure is verified.

We finally consider inverses. $\overline{0}+\overline{0}=\overline{0+0}=\overline{0}$, thus $\overline{0}$ is its own inverse. $\overline{1}+\overline{2}=\overline{3}=\overline{0}$, so $\overline{1}$ is the inverse of $\overline{2}$. Thus, inverses are also in the set. So, we have verified that $\mathbb{Z}/\mathbb{Z}3$ is a subgroup of $\mathbb{Z}$ (actually $\mathbb{Z}^+$ to be precise).

We now return to the set of congruence classes $\mathbb{F}_p$ mentioned earlier. This is defined as

$$\mathbb{F}_p=\left\{\overline{0},\overline{1},\ldots,\overline{p-1}\right\}=\mathbb{Z}/\mathbb{Z}p$$

where p is a prime integer. This set of congruence classes is actually a *field*, called a *prime field*. The concept of fields will be discussed in a later section. Briefly, fields are similar to rings except they are also closed under multiplicative inverses for non-zero elements.

With this background we look to show that $\phi:\mathbb{Z}\longrightarrow\mathbb{F}_p$ is a ring homomorphism.

Firstly, we check that $\phi(1_R)=1_{R'}$ where $R=\mathbb{Z}$ and $R'=\mathbb{F}_p$. If we map 1 to the congruence class $\overline{1}$ and we recall from the definition $\overline{a}=\{\ldots,a-n,a,a+n,a+2n,\ldots\}$ or $\overline{1}=\{\ldots,1-n,1,1+n,1+2n,\ldots\}$ where n is some fixed positive integer, we see that $\phi(1_R)=1_{R'}$, where $1_R=1$ and $1_{R'}=\overline{1}$.

We next verify $\phi(a+b)=\phi(a)+\phi(b)$. Suppose a and b are any two integers in $\mathbb{Z}$. Then under ϕ we have $\phi(a+b)=\overline{a+b}=\overline{a}+\overline{b}$ from the definition of the sum of congruence classes given previously. Then finally we have $\overline{a}+\overline{b}=\phi(a)+\phi(b)$, which completes the verification, this last result coming from the definition of ϕ.

Finally, we check $\phi(ab)=\phi(a)\phi(b)$. Again, suppose a and b are any two integers in $\mathbb{Z}$. Then we have $\phi(ab)=\overline{ab}=\overline{a}\overline{b}$ from the definition of multiplication for congruence classes. Then finally we obtain $\overline{a}\overline{b}=\phi(a)\phi(b)$ completing the verification, where again this last step comes from the definition of ϕ. This completes the solution to the exercise.

Let us now consider another proof exercise regarding ring homomorphisms.

Exercise Prove the "Substitution Principle" given in Artin as Proposition 11.3.4. Namely, suppose $\phi : R \longrightarrow R'$ is a ring homomorphism, and let $R[x]$ be the ring of polynomials with coefficients in R. Let α be an element of R'. Prove there is a unique homomorphism $\Phi : R[x] \longrightarrow R'$ that agrees with the map ϕ on constant polynomials, and that sends $x \rightsquigarrow \alpha$.

Solution We proceed using proof by contradiction. Suppose Φ is not unique. Let Φ' be a distinct homomorphism that also agrees with the map ϕ on constant polynomials, and that sends $x \rightsquigarrow \alpha$. As a reminder, constant polynomials are those with the highest degree power of the terms x^n having a non-zero coefficient being $n = 0$.

Let's construct a general element of $R[x]$, say

$$b(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$$

where a_i are elements of R. Next, let's consider the application of Φ to $b(x)$, namely

$$\Phi(b(x)) = \Phi(a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0)$$

Since Φ is a ring homomorphism, we may apply the properties to yield

$$\Phi(b(x)) = \Phi(a_n)\Phi(x^n) + \Phi(a_{n-1})\Phi(x^{n-1}) + \ldots + \Phi(a_1)\Phi(x) + \Phi(a_0)$$

Since we are told that Φ agrees with ϕ for constant polynomials and it sends $x \rightsquigarrow \alpha$, we can rewrite the above as

$$\Phi(b(x)) = \phi(a_n)\alpha^n + \phi(a_{n-1})\alpha^{n-1} + \ldots + \phi(a_1)\alpha + \phi(a_0)$$

Now, suppose we apply the assumed distinct homomorphism Φ' to $b(x)$, we obtain in a similar fashion to earlier

$$\Phi'(b(x)) = \Phi'(a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0)$$
$$\Phi'(b(x)) = \Phi'(a_n)\Phi'(x^n) + \Phi'(a_{n-1})\Phi'(x^{n-1}) + \ldots + \Phi'(a_1)\Phi'(x) + \Phi'(a_0)$$

We have assumed that Φ' agrees with ϕ for constant polynomials and sends $x \rightsquigarrow \alpha$, but is otherwise distinct. Thus, applying these conditions to the above yields

$$\Phi'(b(x)) = \phi(a_n)\alpha^n + \phi(a_{n-1})\alpha^{n-1} + \ldots + \phi(a_1)\alpha + \phi(a_0)$$

We can therefore see that $\Phi'(b(x)) = \Phi(b(x))$ which is a contradiction of our assumption. Therefore, we conclude that Φ is unique.

We now move onto the concept of an ideal of a ring, with the definition from Artin below.

Definition An *ideal* I of a ring R is a nonempty subset of R with these properties:

- I is closed under addition.
- If s is in I and r is in R, then rs is in I.

Let us explore ideals in the following proof exercise, adapted from Artin.

Exercise Prove the kernel K of the ring homomorphism $\phi : \mathbb{Z}[x] \longrightarrow \mathbb{F}_p$ that sends $f(x)$ to the residue of $f(0) \mod p$ is the ideal (p, x) of $\mathbb{Z}[x]$ generated by p and x.

Solution To begin with, let us clarify some of the terminology.

The kernel of a ring homomorphism is the set of elements that map to zero. To be precise (following Artin), suppose we have the ring homomorphism $\phi : R \longrightarrow R'$, where R and R' are rings. Then the kernel of ϕ is given as

$$\ker \phi = \{s \in R | \phi(s) = 0\}$$

The next term is the residue of $f(x) \mod p$ denoted as $\overline{f}(x)$. This can be best described by the following map

$$f(x) = a_n x^n + \cdots + a_0 \rightsquigarrow \overline{a}_n x^n + \cdots + \overline{a}_0 = \overline{f}(x)$$

where $\overline{a}_i$ is the congruence class of the integer a_i, as we discussed earlier. We note that p is a prime in reference to $\mathbb{F}_p$.

Lastly, the ideal (p, x) is the set of elements defined as $(p, x) = \{r_1 x + r_2 p | r_i \in \mathbb{Z}\}$. We note that both p and x are elements of the ring $\mathbb{Z}$.

Recalling the definition of $\mathbb{F}_p$

$$\mathbb{F}_p = \left\{\overline{0}, \overline{1}, \ldots, \overline{p-1}\right\} = \mathbb{Z}/\mathbb{Z}p$$

where for a given integer a, we have $\overline{a}$ defined by

$$\overline{a} = \{\ldots, a - n, a, a + n, a + 2n, \ldots\}$$

and $n = p$ in this case.

We see that p is certainly in the kernel, since p maps to $\overline{0}$ which is the zero element of $\mathbb{F}_p$. Secondly, since $f(x)$ is sent to $\overline{f}(0)$ this means that

$$f(x) = a_n x^n + \cdots + a_0 \rightsquigarrow \overline{a}_n 0^n + \cdots + \overline{a}_0 = \overline{f}(0)$$

so that $\overline{f}(0) = 0$ whenever $a_0 = 0 \mod p$. Thus both p and x are in the kernel. Since the mapping is a ring homomorphism, we know that mapping the ideal $\phi(r_1 x + r_2 p) = \phi(r_1 x) + \phi(r_2 p) = \phi(r_1)\phi(x) + \phi(r_2)\phi(p)$ which is $\phi(r_1)0 + \phi(r_2)0 = 0$ based on what was discussed earlier. Thus, the ideal is in the kernel of the ring homomorphism.

We now discuss uniqueness of the ideal as the kernel. Clearly, any other non-zero constant added to the ideal will not map to $\overline{0}$. Furthermore, any element of $f(x)$ can be obtained from a multiple of the ideal x by another factor x^{n-1} with a coefficient r_n.

We briefly introduce a new term before we move onto the next section.

Definition (From Artin) A subset V of complex n-space $\mathbb{C}^n$ that is the set of common zeros of a finite number of polynomials in n variables is called an *algebriac variety*, or just *variety*.

(From Artin) An example of a variety is given by a complex line in the (x, y)-plane $\mathbb{C}^2$ (i.e. the usual complex plane). This variety is given by the equation $ax + by + c = 0$.

There is a relationship between ideals and varieties which we examine in the following exercise proof (Adapted from Artin).

Exercise Let I be the ideal of $\mathbb{C}[x_1, \ldots, x_n]$ generated by some polynomials $f_1, \ldots, f_r$, and let R be the quotient ring $\mathbb{C}[x_1, \ldots, x_n]/I$. Let V be the variety of (common) zeros of the polynomials $f_1, \ldots, f_r$ in $\mathbb{C}^n$. Show that the maximal ideals of R are in bijective correspondence with the points of V.

Solution Firstly, let us explain the terminology in the exercise. $\mathbb{C}[x_1, \ldots, x_n]$ is a polynomial ring, with the coefficients from $\mathbb{C}$, i.e. complex numbers. An example for $n = 4$ might be $a_2 x_1^2 x_2 + a_1 x_1 x_4 + a_0$, where the a_i are complex along with the x_i.

Now, we describe quotient rings (from Artin). Let I be an ideal of a ring R. The cosets of the additive subgroup of R are the subsets $a + I$ where $a \in R$. The sets of these cosets are called a *quotient ring*, denoted by R/I. For example, we know that $\{0\}$ is an ideal of Z because $0 + 0$ is zero and thus in the ideal, and any multiple of 0 is also zero and thus in the ideal. Thus, $\mathbb{Z}/\{0\}$ is the quotient ring with elements $\{\ldots, \{-1\}, \{0\}, \{1\}, \{2\}, \ldots\}$.

We should consider what an example might look like of a generated ideal. Such an ideal would have the form $I = a_1 f_1 + a_2 f_2 + \ldots + a_r f_r$, where the a_i are complex. We should be reminded that the ideal itself is a non-empty subset of the original ring R.

Now let us consider maximal ideals. Following Artin, we define a maximal ideal M of a ring R is an ideal that isn't equal to R, and that isn't contained in any ideal other than M and R: If an ideal I contains M, then $I = M$ or $I = R$.

As an example, let us consider the ring of integers $\mathbb{Z}$. Let us consider the principle ideal I, which is all multiples of 2, that is, the set consisting of integers of the form $2n$, where $n \in \mathbb{Z}$. We briefly confirm that this is indeed an ideal. Firstly, we need to confirm it is closed under addition. Suppose two members of our candidate ideal have the form $2n$ and $2m$ with $m, n \in \mathbb{Z}$. The sum is then $2n + 2m = 2(n + m)$. This is of the form $2p$, where $p = n + m$, and thus $p \in \mathbb{Z}$ by the usual sum property integers. Thus, the candidate is closed under addition.

Next, we need to check that the multiplication of an element of the R with an element of the candidate ideal is indeed in the ideal. Suppose we take a typical member of the candidate ideal $2n, n \in \mathbb{Z}$ and an element $r \in \mathbb{Z}$, and form the product $r2n$. We then have $r2n = 2rn = 2p$ by the usual commutativity of multiplication of integers. Since rn is also just an element $p \in \mathbb{Z}$, we have verified the second property of ideals. Thus, the even integers are an ideal of the entire set of integers.

To verify the maximal properties, we first confirm that $I \neq \mathbb{Z}$. Clearly, $3 \in \mathbb{Z}$, but $3 \notin I$, since 3 is an odd integer. Therefore, $I \neq \mathbb{Z}$.

Now, we show that there is no ideal in R which contains I but is not R. Since I is all the even integers, it has the form $2n$ for all $n \in \mathbb{Z}$, as previously noted. If there is an ideal that contains I, which is not I, it certainly must contain at least one odd integer, say $2m + 1$ where m is an arbitrary but fixed member of $\mathbb{Z}$. To be an ideal, this set must be closed under addition. That means all elements of the form $2n+2m+1$ must be contained in the ideal. This can be written as $2(n + m) + 1$ which is odd, and in fact since n takes on all integers, this means that all of the odd integers must be included. Thus, this candidate ideal would have to be R itself since it includes both the even and odd integers. Therefore, we conclude that I, the even integers, is maximal.

For an illustration, let us consider the odd integers briefly. One might suspect this set is a maximal ideal. However, it is not even an ideal. To see this, consider two arbitrary elements $p = 2n+1$ and $q = 2m+1$, where $n, m \in \mathbb{Z}$. We know that to be an ideal, the set must be closed under addition. We compute $p+q = 2n+1+2m+1 = 2n+2m+2 = 2(n+m+1)$ and see it is in the form $2t$, where $t = n+m+1$. Thus, the set is not closed under addition, and thus is not even an ideal, let alone a maximal ideal.

To show bijective correspondence, first we have to show the each point of V is the image of some ideal of R. Recall that a point of V is just a finite set of n complex numbers from the space $\mathbb{C}^n$. In fact, the correspondence is with maximal ideals so we cannot include R itself. Furthermore, R is itself a quotient ring.

Earlier we showed that an example ideal of $\mathbb{C}[x_1, \ldots, x_n]$ is of the form $I = a_1 f_1 + a_2 f_2 + \ldots + a_r f_r$, and this will be the zero ideal for any point in V since all the polynomials are zero. We know that when a ring R has the zero ideal $I = \{0\}$ in the quotient, i.e. $R/I = R/\{0\}$ it returns the original ring itself. This is because we are merely adding the zero element, which is the additive identity, to arbitrary elements of the ring R. However, the entire ring itself is not a maximal ideal since by definition, a maximal ideal of a ring cannot be equal to the ring itself.

Let us consider what the form of the quotient ring would be for a typical ideal. Suppose we have a typical element from $\mathbb{C}[x_1, \ldots, x_n]$, say $a_m x_n^{p_{m,n}} x_{n-1}^{p_{m,n-1}} \cdots x_1^{p_{m,1}} + a_{m-1} x_n^{p_{m-1,n}} x_{n-1}^{p_{m-1,n-1}} \cdots x_1^{p_{m-1,1}} + \ldots + a_0$. Then the quotient ring would be that element added to a given ideal, the form of which might be $b_1 f_1 + b_2 f_2 + \ldots + b_r f_r$, where the b_i are complex.

Now, an ideal of the quotient ring would be a subset of elements formed as described above. Suppose that there were no common zeros of the polynomials f_i. This means that at least one of the terms in $b_1 f_1 + b_2 f_2 + \ldots + b_r f_r$ would be non-zero. Let us show that in this case, a maximal ideal doesn't exist for the quotient ring R. First, we should check to see if an ideal even exists, let alone a maximal.

Certainly, for an ideal of the quotient ring R, it must be closed under addition. So let us see if any subset of R has such a property. Clearly, any element of the original ideal will satisfy this requirement since they are sums of polynomials. Recall that the quotient ring is of the form $a + I$ where $a \in \mathbb{C}[x_1, \ldots, x_n]$ and $I = a_1 f_1 + a_2 f_2 + \ldots + a_r f_r$.

Now, let's check the multiplication property. Let s be one element of the ideal, and r be another one. Then we require rs also be in the ideal. We can see that products in general will not be in the ideal, since all the polynomials occur in the first power. For example, suppose $s = a_1 f_1$ and $r = a_2 f_2$. Then $rs = a_1 a_2 f_1 f_2$ which may not be present in the ideal, unless the product term appears in the ideal. If it happens to be, we can form another product which is not in the ideal. If we have n polynomials, there will be n^2 products, thus by the pigeonhole principle, some product will not be represented. However, this simple conclusion is at odds with the definition of ideal so we remind ourselves that products are closed under the multiplication property of ideals.

If we want to form this subset of the quotient ring consisting of the original ideal, it must be only for $a = 0$. Then, we can argue that this ideal must be not be equal to the entire quotient ring as we assume that any non-zero element $a \in R$ is not included in the ideal in general. This means that it is a candidate maximal ideal. Now, we need to see if there is some ideal other than R that contains I, and is not I.

Let us suppose the form of this candidate ideal is simply $a + I$ where $a \neq 0$ and $a \neq I$. We need to verify this satisfies the properties of an ideal. Let us consider the first property of additive closure by taking a general element of the candidate ideal to be $a' + I'$. Then we have $a'_1 + I'_1 + a'_2 + I'_2$ as a representative sum. We suppose that the a' only differ by a complex coefficient that scales the term. Then, we can write $a'_1 = c'_1 b'$ and $a'_2 = c'_2 b'$. Forming the sum, we have $(c'_1 + c'_2)b'$. Now, as elements from I' are polynomials, sums of these elements will also be polynomials. Therefore, the final result will also be of the form $a' + I'$ or $(c'_1 + c'_2)b' + I'$ to be more explicit. That is to say, closed under addition.

Next we consider the multiplicative property. Recall, this means that if s is in the ideal, and r is in the ring, then rs is in the ideal. This means that $s = a + I$ for some a where $a \neq 0$ and $a \neq I$ as above. We also recall a typical element of the quotient ring is also of the form $r = a + I$. Let us write this as $a' + I'$ to distinguish it from the specific element in the provisional maximal ideal. Then we can form the product $rs = (a' + I')(a + I)$. Assuming there is no restriction on the expansion we obtain $rs = a'a + a'I + I'a + I'I$.

We know that the ideal I is closed under multiplication by definition. We can also say that $a'I, I'a, I'I$ are also closed since they must be elements in the ideal I. The final term $a'a$ is some element in the ring, since a ring is closed under multiplication. Now the question is whether or not $a'a = 0$ or $a'a \in I$. We know that $a, a' \neq 0$ by construction. Now, we ask can two non-zero elements multiply to obtain the zero ring element? If we consider as an example the ring of integers, clearly two non-zero elements will not multiply to obtain the zero element. Now, let us consider each element of the ring, not including elements of the ideal, which are polynomials with complex terms. We can ask the question, is it possible to multiply two such terms and arrive at zero.

We can argue that since each polynomial does not evaluate to zero, then the product of two such polynomials will not evaluate to zero either.

Now, we turn to whether two elements that are not in the ideal I can be multiplied together to obtain a result that is not in the ideal also. Let us look at the definition of ideal for some clues on how to potentially approach this problem. We have the property: If s is in I and r is in R, then rs is in I, where I is an ideal and R is the ring. We might write this symbolically thus:

$$(s \in I \wedge r \in R) \Longrightarrow rs \in I$$

We can write this in the contra-positive form:

$$\neg\,(rs \in I) \Longrightarrow \neg\,(s \in I \wedge r \in R)$$

$$(rs \notin I) \Longrightarrow \neg\,(s \in I) \vee \neg\,(r \in R)$$

Let us review the truth table of the implication symbol.

Implication:

A	$\implies$	B
T	T	T
T	F	F
F	T	T
F	T	F

Now, suppose $rs \neq I$ is true. By assumption, we know that $\neg(s \in I)$ is true. Therefore, the right side of the implication

$$(rs \notin I) \implies \neg(s \in I) \vee \neg(r \in R)$$

is true. Now, if the left side was false, the implication would still be true. If we recall the original formulation:

$$(s \in I \wedge r \in R) \implies rs \in I$$

and suppose the left side is false since $s \notin I$, then we see that the implication is true when the right hand side is true or false. Thus, we see that $s \notin I$ is not sufficient.

We might consider opportunities to potentially constrain the situation. Is it possible to construct the following conditions: Suppose $r, s \in R$ and $r \notin I$ and $s \notin I$. Let us suppose that there is another ideal, J, in R, where $I \cap J = \emptyset$. Let us further suppose that $r, s \in J$. Now, let us consider $J + I$ to be the form of the candidate ideal. That, the candidate ideal is the sum of two "smaller" ideals. Let us see if this is a legitimate formulation.

We first should verify the properties of an ideal are satisfied. First, we check closure under addition. Let us refer to $I+J$ as K for convenience. Let $r \in I$ and $s \in J$. Further, let $r' \in I$. We see that $r, r' \in K$. Then certainly, $r+r' \in I \in K$. This is because I as an ideal is closed under addition. Likewise, let us consider $s, s' \in J$ and compute the sum $s + s'$. By closure of ideals under addition, this sum will also be in J. Next let us consider the various products.

We take $r \in I$ and $s \in J$. Now we form the product. We compute rs and consider the properties of ideals. If s is in I and r is in R, then rs is in I, where I is the ideal and R is the ring.

3.3 Fields

3.4 Vector Spaces

3.5 Transformations

3.6 Representation Theory

3.7 Modules

3.8 Galois Theory

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Chapter 5

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